

LAB INTERNET OF THINGS

INSTRUMENTATION AND COMPUTATION RESEARCH GROUP

**DESIGN ENGINEERING INTEGRASI
MULTI-MODALITY
GAS LEAK DETECTOR DENGAN
THERMAL DAN FIELD TESTING DI
UNIT KILANG**

PROJECT REPORT JUNE 2026



PERSIAPAN FIELD TESTING

Gas Leak Sensor Unit

- Uji Komponen PCB GLD v1 (done)
- Uji Komponen PCB GLD v2 (done)
- Uji Battery GLD (progress)
- Assembly Casing GLD, Battery (sirine and buzzer masih pengiriman dari china)
- Desain Gas Test Chamber (done)
- Assembly Gas Test Chamber - mekanik & electrical (progress)
- Pengumpulan Dataset GLD (progress)
- Development Model ML/AI (saat ini model yang dibangun baru untuk LPG - perlu konfirmasi gas apa saja yang perlu di latih untuk di deteksi)
- Pengujian & Validasi Model ML/AI (progress)
- Uji Fungsional Sistem [GLD-CH-GW-SERVER] (progress)

GAS LEAK SENSOR UNIT

No	Workstream	Activity	Version / Scope	Status	Remarks
1	PCB Development	Component testing for PCB GLD	GLD v1	Done	Completed and verified
2	PCB Development	Component testing for PCB GLD	GLD v2	Done	Completed and verified
3	Power System Testing	GLD battery testing	Battery performance validation	Done	Battery performance test completed
4	Mechanical Assembly	GLD casing and battery assembly	Sensor unit enclosure	In Progress	Siren and buzzer components are still in
5	Test Equipment	Gas Test Chamber design	Chamber design	Done	Design completed
6	Test Equipment	Gas Test Chamber assembly	Mechanical & electrical assembly	In Progress	Ongoing fabrication and integration
7	Data Collection	GLD dataset collection	Sensor response dataset	In Progress	Ongoing data acquisition
8	AI/ML Development	ML/AI model development	LPG detection model	In Progress	Current model is focused on LPG; additional target gas
9	AI/ML Validation	ML/AI model testing and validation	Detection accuracy validation	In Progress	Ongoing validation
10	System Integration	Functional system testing	GLD-CH-GW-Server	In Progress	End-to-end system validation in progress

PERSIAPAN FIELD TESTING

Cluster Head / Lora Mesh Network

- Uji Komponen PCB CH v1 (done)
- Uji Komponen PCB CH v2 (done)
- Uji Komponen PCB CH v3 (progress)
- Desain & Development Casing CH v1 (done)
- Desain & Development Casing CH v2 (done)
- Desain & Development Casing CH v3 (progress)
- Desain Tiang CH 6 meter + Bracket Solar Panel (done - to be discuss dengan RU Pertamina)
- Desain & Development LoRa Tester (done)
- Uji Jarak & Reliabilitas Komunikasi Data LoRa (progress)
- Uji Arsitektur Jaringan Mesh (progress)
- Uji Konsumsi Solar Panel & Baterai CH (done)

Cluster Head / Lora Mesh Network

No	Workstream	Activity	Version / Scope	Status	Remarks
1	PCB Development	Component testing for PCB Cluster Head	CH v1	Done	Completed and verified
2	PCB Development	Component testing for PCB Cluster Head	CH v2	Done	Completed and verified
3	PCB Development	Component testing for PCB Cluster Head	CH v3	In Progress	Ongoing validation
4	Mechanical Design	Casing design & development	CH v1	Done	Completed
5	Mechanical Design	Casing design & development	CH v2	Done	Completed
6	Mechanical Design	Casing design & development	CH v3	In Progress	Under development
7	Installation Support	6-meter CH pole design with solar panel bracket	Field installation support	Done	To be discussed with RU Pertamina
8	Testing Tools	LoRa Tester design & development	Field testing tool	Done	Ready for testing activity
9	Communication Testing	LoRa data communication range & reliability test	Field testing preparation	In Progress	Ongoing measurement
10	Network Validation	Mesh network architecture test	LoRa mesh topology	In Progress	Ongoing validation
11	Power System Testing	Solar panel and CH battery consumption test	Power endurance validation	Done	Completed

PERSIAPAN FIELD TESTING

Server Application

- Development Mock up Dashboard Monitoring (done - to be discussed dengan pertamina)
- Development Backend Server & Database (progress)

No.	Modul	Aktivitas	Status	Keterangan
1	Server Application	Development <i>Mock-up Dashboard Monitoring</i>	✅ Done	Siap untuk didiskusikan dan direview bersama Pertamina.
2	Server Application	Development <i>Backend Server & Database</i>	🔄 In Progress	Pengembangan backend, API, dan database masih berlangsung.

PERSIAPAN FIELD TESTING

Kesimpulan Persiapan Field Testing

Berdasarkan hasil pengembangan sistem dan kesiapan perangkat, proyek telah mencapai tingkat kesiapan yang memadai untuk melaksanakan site survey awal di RU IV Cilacap. Tahap ini belum ditujukan untuk kegiatan instalasi, melainkan sebagai studi lapangan untuk memvalidasi desain jaringan LoRa Mesh, melakukan pemetaan lokasi pemasangan perangkat, serta mengevaluasi kondisi lingkungan yang memengaruhi kualitas komunikasi.

Site survey juga bertujuan untuk mengonfirmasi pembagian ruang lingkup pekerjaan (scope of work) antara LGU/ITB dan RU Pertamina, termasuk kebutuhan infrastruktur, utilitas, dan material pendukung sebagaimana tercantum dalam dokumen persiapan proyek. Hasil survey akan menjadi dasar penyempurnaan Detail Engineering Design (DED), finalisasi jumlah dan lokasi perangkat, serta penentuan kesiapan instalasi pada tahap implementasi berikutnya.

GAS LEAK SENSOR UNIT

UJI KOMPONEN PCB GLD

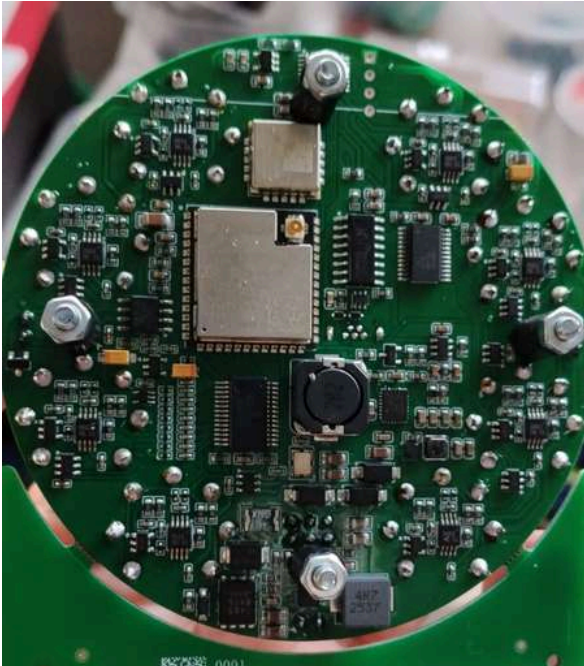
```
--- I2C Scan (direct bus) ---  
0x71 found  
Total: 1 device(s)
```

```
--- I2C Scan per TCA9548 channel ---  
ch0: 0x60  
ch1: 0x60  
ch2: 0x60  
ch3: 0x60  
ch4: 0x60  
ch5: 0x60  
ch6: 0x60  
ch7: 0x60
```

```
--- ADS1256 All Channels ---  
AIN0: raw= 2796248 1.666694 V  
AIN1: raw= 3600133 2.145846 V  
AIN2: raw= 2934370 1.749021 V  
AIN3: raw= 2568856 1.531157 V  
AIN4: raw= 2408928 1.435833 V  
AIN5: raw= 2363616 1.408825 V  
AIN6: raw= 2872853 1.712354 V  
AIN7: raw= 2942468 1.753848 V
```

```
[5] MCP4725 @ 0x60... ch0: MCP4725 FOUND  
ch1: MCP4725 FOUND  
ch2: MCP4725 FOUND  
ch3: MCP4725 FOUND  
ch4: MCP4725 FOUND  
ch5: MCP4725 FOUND  
ch6: MCP4725 FOUND  
ch7: MCP4725 FOUND
```

```
[6] ADS1256 SPI... 10ms  
ADS1256 SPI OK 10ms
```



Hasil: Output Board GLD Baru semua parameter OK!

Desain hardware node baru telah berhasil diverifikasi. Semua output modul terkonfirmasi berjalan normal.

PSU 5V 10A -> Circuit 5V Board (5.05V)
Tegangan VCC ESP32 = 3.305V
VSYN : 2,573V
VBUS : 4.97V

UJI KOMPONEN PCB GLD

```
-----  
[LOOP #910 | 00:15:15]  
-----  
[ADC CH0] 1979706 | 0.009200 V | GAIN x64  
[ADC CH1] 1866987 | 0.008676 V | GAIN x64  
[ADC CH2] 1845150 | 0.008575 V | GAIN x64  
[ADC CH3] 1549614 | 0.007202 V | GAIN x64  
[ADC CH4] 1833502 | 0.008521 V | GAIN x64  
[ADC CH5] 1728211 | 0.008032 V | GAIN x64  
[ADC CH6] 1936471 | 0.008999 V | GAIN x64  
[ADC CH7] 1845221 | 0.008575 V | GAIN x64  
[LORA] -  
[SYS] TCA=1 ADS=1 LORA=1 MCP=8/8  
-----
```

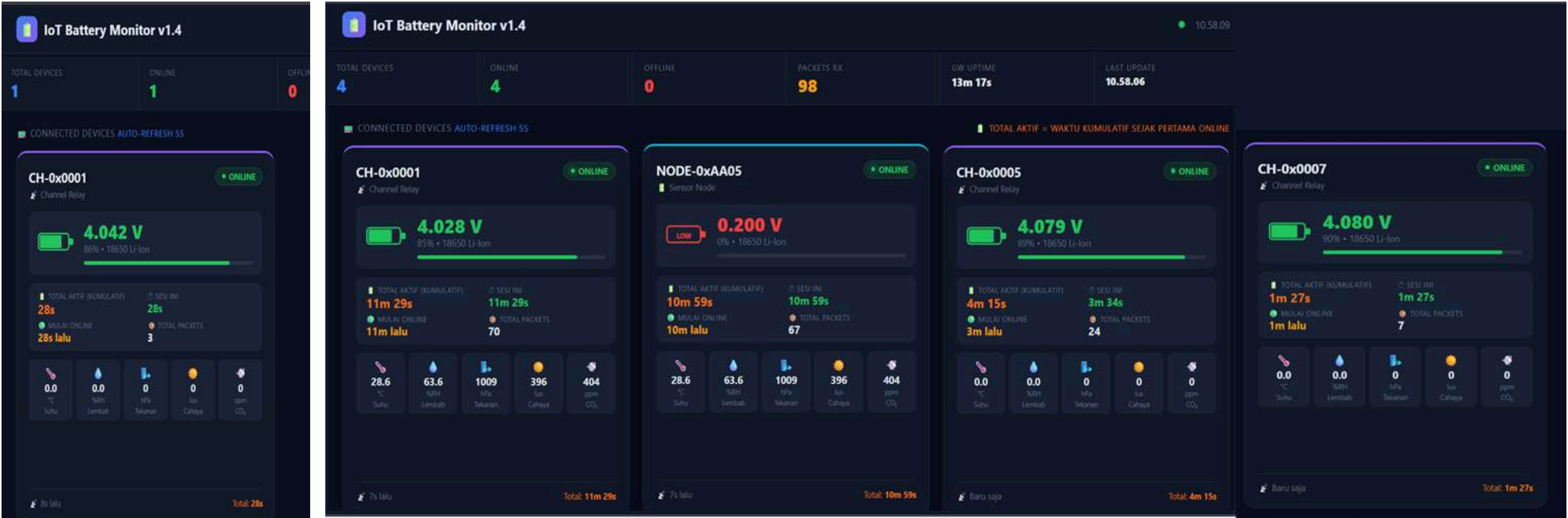
Terminal Output – Loop #910 | 00:15:15

Verifikasi parameter ADC, LoRa, dan sensor pada board baru

TCA TCA9548A I2C Multiplexer	1 ✓
ADS ADS1115 ADC 16-bit	1 ✓
MCP MCP3208 – 8 Channel ADC OK	8/8 ✓
ADC CH0–CH7 aktif Tegangan CH0: 0.009V · Gain x64 · Semua channel terbaca normal	

UJI KOMPONEN PCB GLD

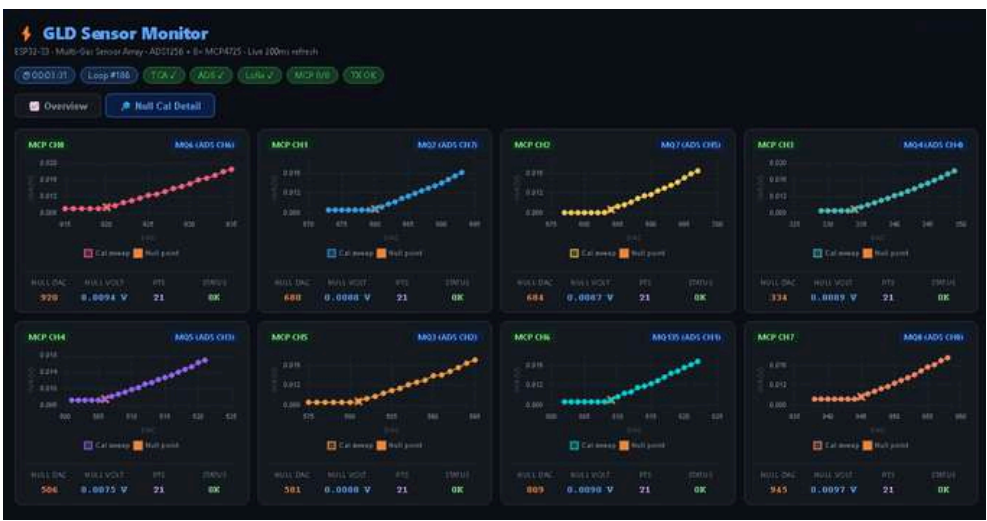
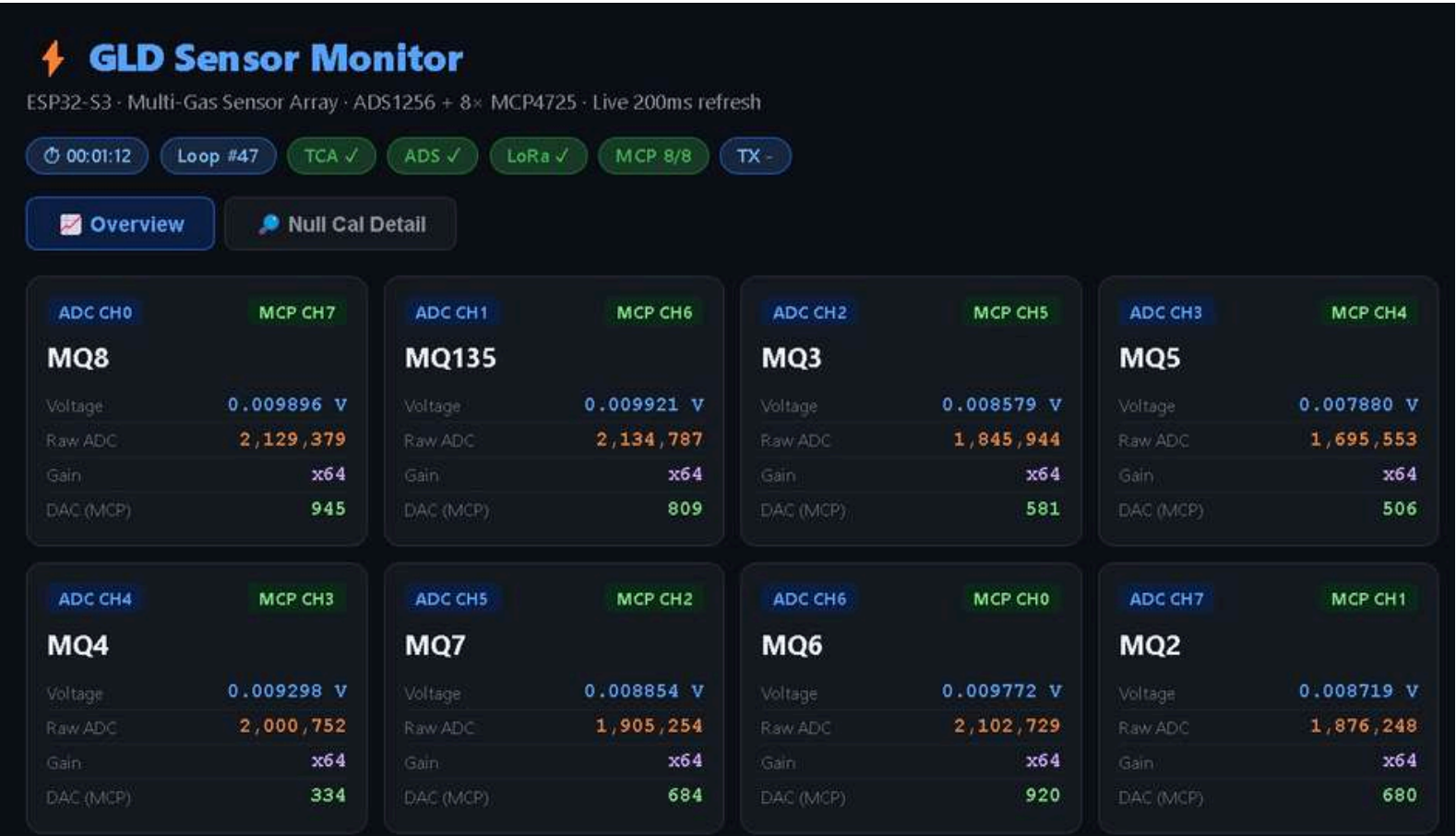
Uji Konsumsi Baterai GLD



CH-0x0001	4.028 V · 85%	ONLINE
NODE-0xAA05	0.200 V · 0%	LOW BATT
CH-0x0005	4.079 V · 89%	ONLINE
CH-0x0007	4.080 V · 90%	ONLINE

UJI KOMPONEN PCB GLD

Pengecekan pembacaan masing-masing sensor MQ pada PCB Board GLD untuk memastikan seluruh sensor berfungsi dan menghasilkan data sesuai spesifikasi.



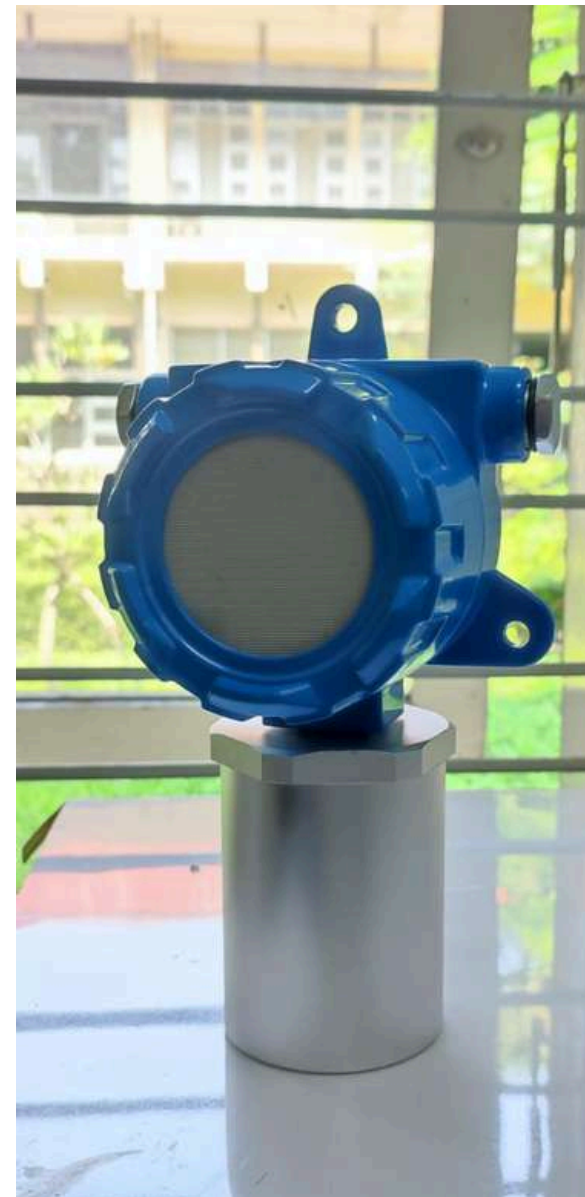
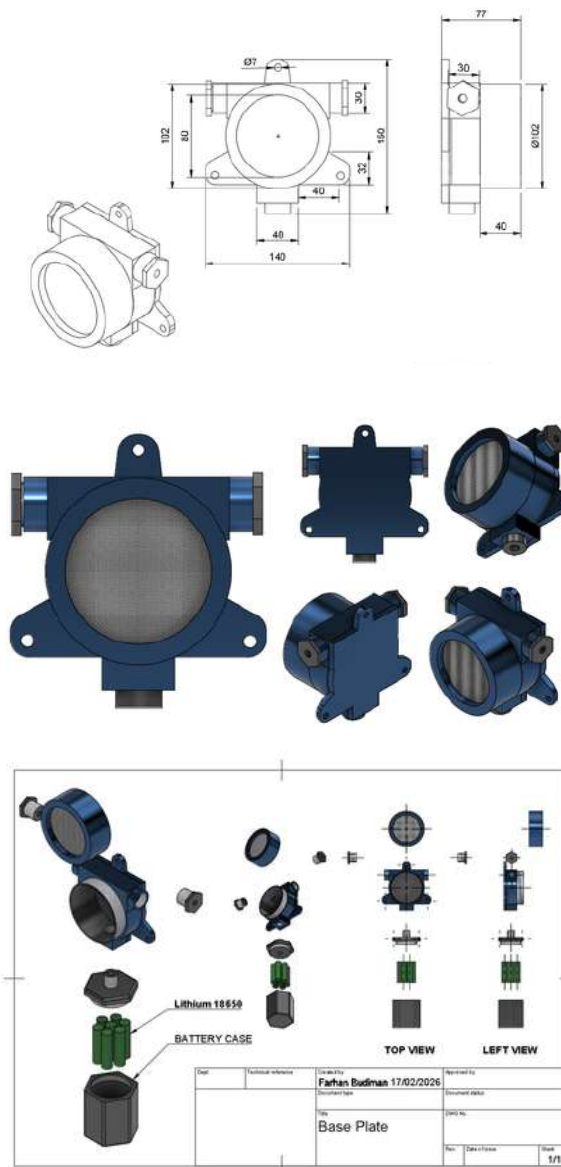


proses QC dan Warming up sensor MQ

+ ≡ Batch 1 ▾ Batch 2 ▾

PROGRESS GAS LEAK SENSOR & AI

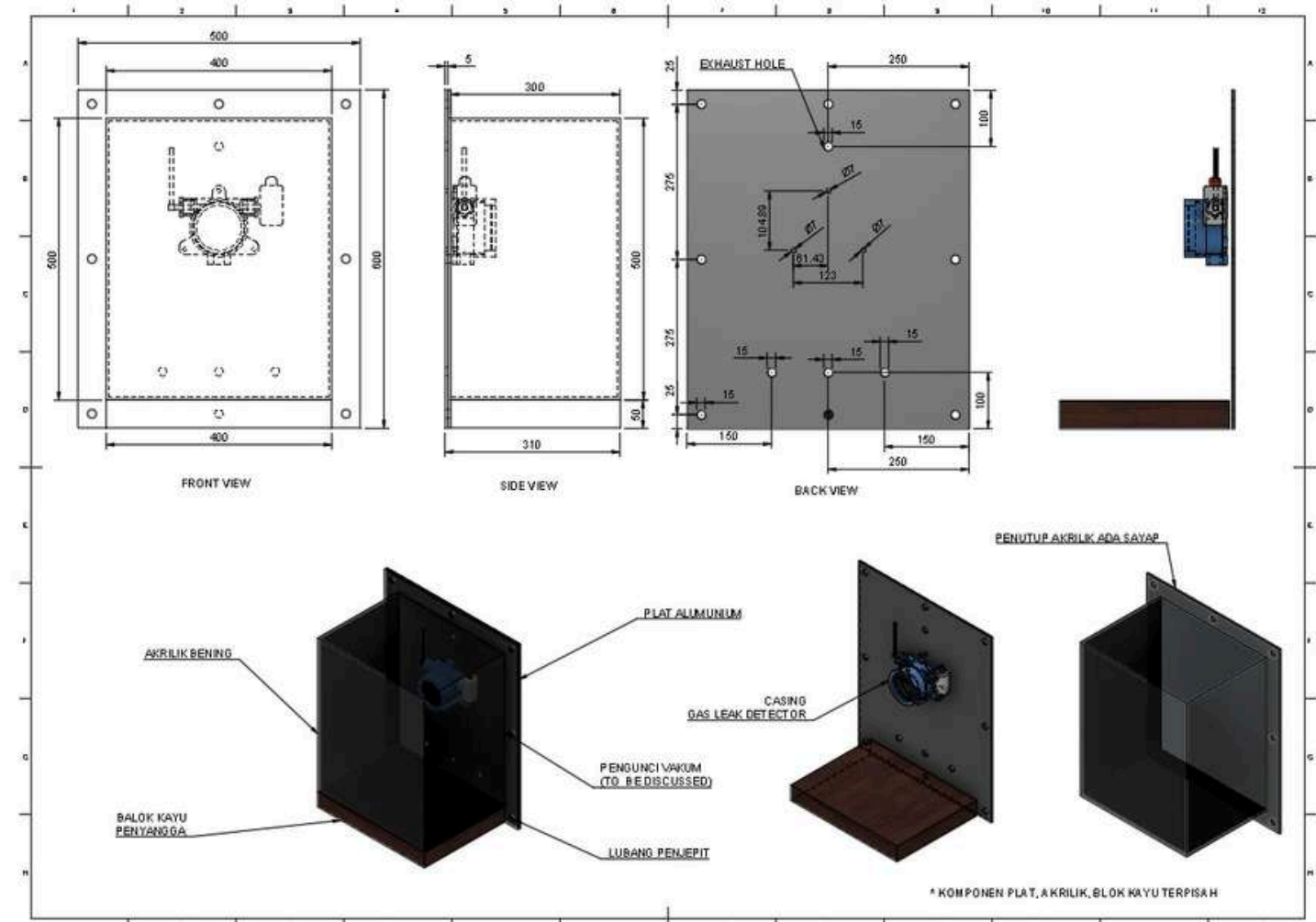
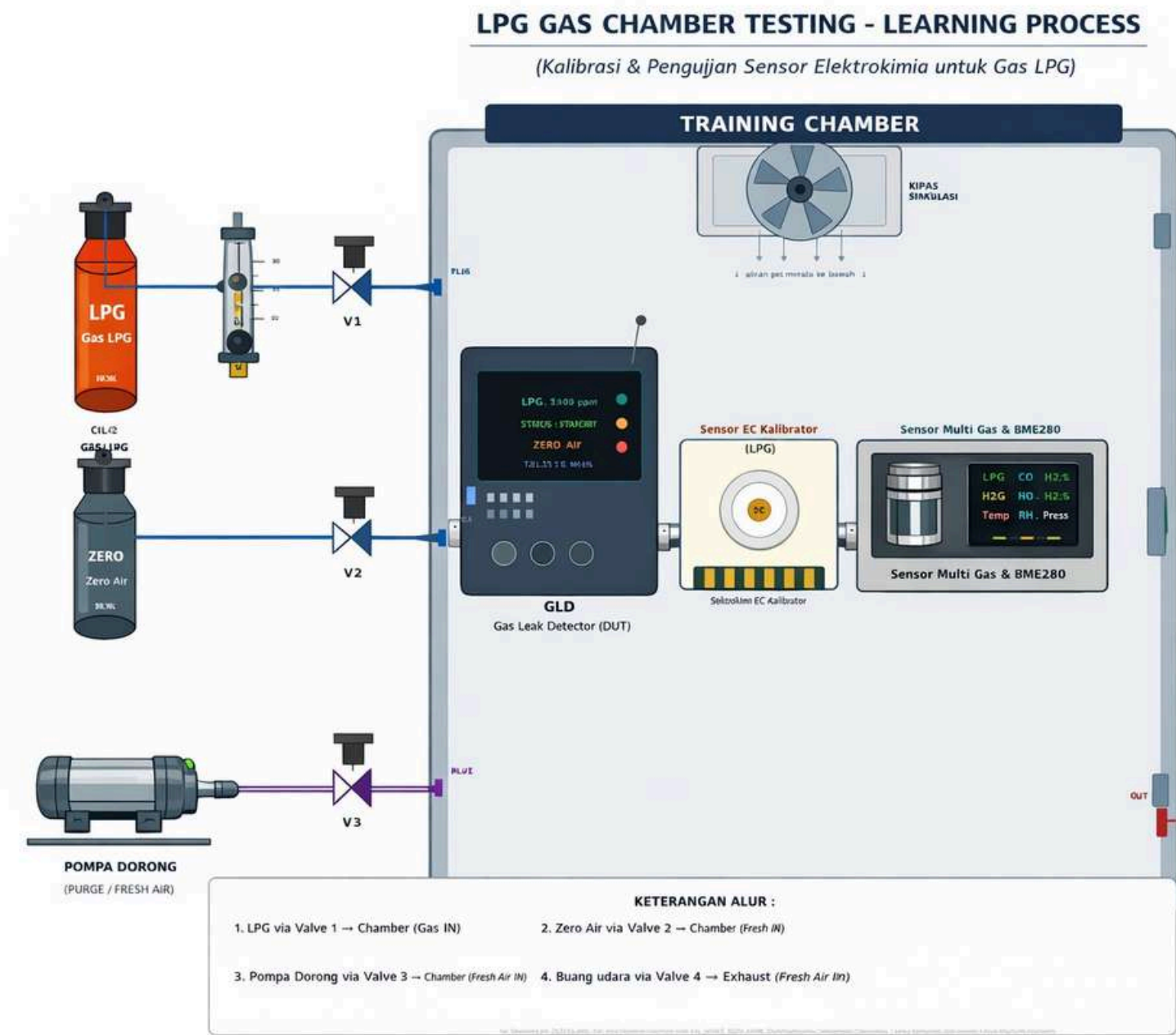
Casing GLD telah selesai dirakit (assembly) di Laboratorium Fisika ITB sebagai persiapan untuk tahap pengujian dan implementasi.



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DESAIN GAS TEST CHAMBER



Assembly Gas Test Chamber



Pengambilan Dataset Baru untuk Model ML GLD

Pengambilan dataset baru telah dilakukan menggunakan PCB GLD versi terbaru sebagai dasar pengembangan dan peningkatan akurasi model Machine Learning (ML). Dataset yang diperoleh akan digunakan untuk proses optimasi model GLD yang akan diimplementasikan di area kilang, sehingga model memiliki performa yang lebih representatif terhadap kondisi operasional di lapangan.

Captured Dataset Inventory

Raw files in test LPG board 12

File	Rows	Duration
Board12_raw_dataset (1).csv	737	736.0s
Board12_raw_dataset (2).csv	130	129.0s
Board12_raw_dataset (3).csv	270	269.0s
Board12_raw_dataset (4).csv	328	327.0s
Board12_raw_dataset (5).csv	1,881	1880.0s
Board12_raw_dataset.csv	160	159.0s

Berikut ditampilkan aplikasi Dataset Recorder yang digunakan untuk proses akuisisi data beserta hasil dataset yang telah berhasil dikumpulkan sebagai dasar pengembangan dan optimasi model Machine Learning (ML).

Gas Sensor Recorder

not recording | disconnected

Baud115200BoardBoard12

ConnectDisconnectRecordStopExcelCSVClear

Rows0Rate0/s

Air status: waiting for recordingRecord clean air first. Baseline uses first 60s.

MQ135Vwaiting

MQ2Vwaiting

MQ3Vwaiting

MQ4Vwaiting

MQ7Vwaiting

MQ5Vwaiting

MQ6Vwaiting

MQ8Vwaiting

Live MQ Voltages

WindowAllAuto-followLatest

Board12_raw_dataset (5) - Excel

FILEHOMEINSERTPAGE LAYOUTFORMULASDATAREVIEWVIEWACROBAT

Calibri10A

General

Conditional FormattingTable Cell Styles

InsertDeleteFormat

AutoSumFillSort & Find & FilterClear

Adobe Acrobat

sample_id

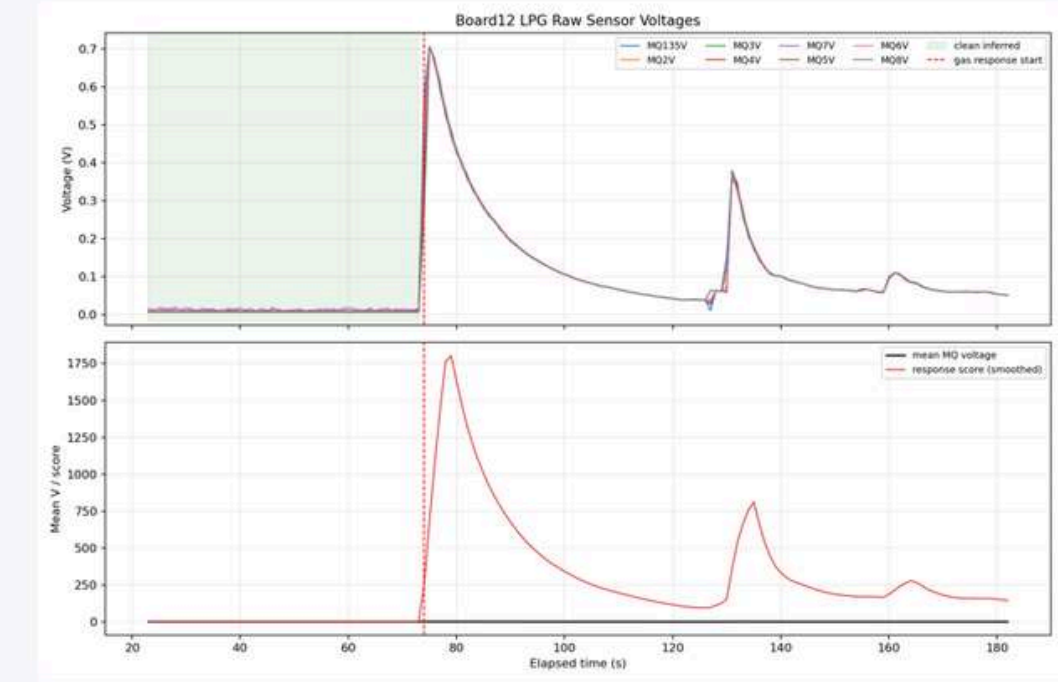
sample_id	recorded_at	elapsed_ms	board_name	board_id	MQ135V	MQ2V	MQ3V	MQ4V	MQ7V	MQ5V	MQ6V	MQ8V
6	2026-05-06T06:50:14.626Z	5005	Board12	12	0.058608	0.059025	0.058518	0.058163	0.058356	0.058619	0.058817	0.058431
7	2026-05-06T06:50:15.626Z	6005	Board12	12	0.057449	0.057935	0.058087	0.058133	0.058098	0.057723	0.057822	0.057994
8	2026-05-06T06:50:16.625Z	7004	Board12	12	0.057779	0.057856	0.057774	0.058283	0.058173	0.058507	0.058126	0.058134
9	2026-05-06T06:50:17.626Z	8005	Board12	12	0.058025	0.058132	0.058055	0.058941	0.058351	0.058114	0.057813	0.057607
10	2026-05-06T06:50:18.627Z	9006	Board12	12	0.058037	0.05841	0.057701	0.058375	0.058099	0.058007	0.058501	0.057988
11	2026-05-06T06:50:19.626Z	10005	Board12	12	0.058001	0.058185	0.057922	0.057876	0.057728	0.058362	0.05802	0.058481
12	2026-05-06T06:50:20.627Z	11006	Board12	12	0.058316	0.057819	0.058264	0.058271	0.057896	0.057678	0.057729	0.058378
13	2026-05-06T06:50:21.626Z	12005	Board12	12	0.05819	0.057797	0.057542	0.058107	0.057449	0.058256	0.058014	0.058231
14	2026-05-06T06:50:22.626Z	13005	Board12	12	0.057429	0.056942	0.057546	0.057522	0.057579	0.057936	0.056875	0.057465
15	2026-05-06T06:50:23.626Z	14005	Board12	12	0.057531	0.057659	0.056812	0.057178	0.057148	0.056708	0.056756	0.056928
16	2026-05-06T06:50:24.624Z	15003	Board12	12	0.057599	0.057155	0.057257	0.057582	0.056773	0.056743	0.057729	0.057314
17	2026-05-06T06:50:25.624Z	16003	Board12	12	0.057302	0.057717	0.057518	0.057114	0.057212	0.05732	0.057529	0.057355
18	2026-05-06T06:50:26.624Z	17003	Board12	12	0.05759	0.056957	0.057064	0.057018	0.056933	0.056428	0.056913	0.057699
19	2026-05-06T06:50:27.625Z	18004	Board12	12	0.057268	0.057399	0.057161	0.057644	0.056987	0.057552	0.057101	0.057507
20	2026-05-06T06:50:28.624Z	19003	Board12	12	0.056793	0.056916	0.057164	0.057221	0.057458	0.057003	0.057583	0.057219
21	2026-05-06T06:50:29.624Z	20003	Board12	12	0.057045	0.057645	0.057562	0.05772	0.057894	0.057629	0.058035	0.057543
22	2026-05-06T06:50:30.624Z	21003	Board12	12	0.057869	0.056943	0.05796	0.057735	0.057292	0.057553	0.057555	0.057397
23	2026-05-06T06:50:31.624Z	22003	Board12	12	0.057698	0.057908	0.057784	0.058331	0.057754	0.058399	0.057383	0.057119
24	2026-05-06T06:50:32.624Z	23003	Board12	12	0.057061	0.057032	0.0576	0.056665	0.056807	0.056982	0.057321	0.057359
25	2026-05-06T06:50:33.624Z	24003	Board12	12	0.056999	0.057228	0.057567	0.057048	0.057498	0.05702	0.057506	0.056879

Board12_raw_dataset (1).csv	06/05/2026 12:03	Microsoft Excel C...	86 KB
Board12_raw_dataset (2).csv	06/05/2026 13:14	Microsoft Excel C...	16 KB
Board12_raw_dataset (3).csv	06/05/2026 13:20	Microsoft Excel C...	32 KB
Board12_raw_dataset (4).csv	06/05/2026 13:40	Microsoft Excel C...	39 KB
Board12_raw_dataset (5).csv	06/05/2026 14:22	Microsoft Excel C...	221 KB
Board12_raw_dataset.csv	06/05/2026 10:37	Microsoft Excel C...	19 KB

Saat ini, sampel gas yang digunakan untuk membangun model Machine Learning (ML) terdiri dari dua kelas, yaitu LPG dan No Gas, sebagai dataset awal untuk proses pelatihan dan validasi model.

Signal Analysis Example

First LPG run shows clean baseline, sharp response, then slow decay



```
Board12 LPG / Liquified Butane Sensitivity Analysis

Captures analyzed: 2
  Capture 1: 'Board12_raw_dataset.csv', 160 rows, 159.0s duration, response start 73.996s, peak 75.0s
  Capture 2: 'Board12_raw_dataset (1).csv', 737 rows, 736.0s duration, response start 188.001s, peak 670.0s

## Aggregate Sensitivity Ranking
Ranked by average peak delta from clean-air baseline across the captures.
1. MQ5V: avg delta 0.65110 V, avg SNR 3255.5, baseline 0.00709 V
2. MQ135V: avg delta 0.65054 V, avg SNR 2180.3, baseline 0.00873 V
3. MQ3V: avg delta 0.65048 V, avg SNR 3252.4, baseline 0.00852 V
4. MQ2V: avg delta 0.64995 V, avg SNR 3249.8, baseline 0.00854 V
5. MQ8V: avg delta 0.64882 V, avg SNR 1962.8, baseline 0.01005 V
6. MQ4V: avg delta 0.64663 V, avg SNR 738.8, baseline 0.01073 V
7. MQ6V: avg delta 0.64587 V, avg SNR 449.2, baseline 0.01141 V
8. MQ7V: avg delta 0.64273 V, avg SNR 193.2, baseline 0.01457 V

Most sensitive by average voltage delta: 'MQ5V' (0.65110 V average peak delta).
Least sensitive by average voltage delta: 'MQ7V' (0.64273 V average peak delta).

## Per-Capture Rankings
### Capture 1: Board12_raw_dataset.csv
1. MQ2V: delta 0.69967 V, SNR 3498.3
2. MQ135V: delta 0.69899 V, SNR 1358.1
3. MQ3V: delta 0.69898 V, SNR 3494.9
4. MQ5V: delta 0.69867 V, SNR 3493.4
5. MQ8V: delta 0.69847 V, SNR 3492.1
6. MQ4V: delta 0.69597 V, SNR 1263.8
7. MQ6V: delta 0.69512 V, SNR 321.6
8. MQ7V: delta 0.69308 V, SNR 226.9

### Capture 2: Board12_raw_dataset (1).csv
1. MQ5V: delta 0.68352 V, SNR 3017.6
2. MQ135V: delta 0.68218 V, SNR 3016.5
3. MQ3V: delta 0.68198 V, SNR 3009.9
4. MQ2V: delta 0.68024 V, SNR 3001.2
5. MQ8V: delta 0.59918 V, SNR 433.3
6. MQ4V: delta 0.59729 V, SNR 214.5
7. MQ6V: delta 0.59662 V, SNR 576.9
8. MQ7V: delta 0.59237 V, SNR 159.6

## Notes
- Sensitivity here means voltage increase from the initial clean-air baseline to the maximum LPG/butane response in the same capture.
- The first capture has a very strong saturation-like spike near 0.70 V on all channels; rankings are therefore close together.
- Use the aggregate ranking cautiously until you collect more repeated runs with slower gas rising and complete clearing.
```


Berdasarkan dataset yang diperoleh dari PCB GLD versi terbaru, model Machine Learning (ML) yang saat ini dikembangkan menggunakan metode Temporal Convolutional Network (TCN).

The image is a presentation slide with a dark blue background and light blue accents. At the top left, there is a light blue rectangular box containing the text 'Lab IoT - ITB'. Below this, the title 'TCN' is written in large, white, serif capital letters. Underneath the title, 'Gas LPG Early Warning' is written in a smaller, light blue, sans-serif font. A descriptive line of text in white, sans-serif font reads: 'Temporal Convolutional Network untuk prediksi kebocoran gas 5 detik sebelum terjadi'. At the bottom of the slide, there are three light blue rectangular boxes arranged horizontally. The first box contains '3,234' in large light blue font and 'data' in smaller white font below it. The second box contains '8' in large light blue font and 'sensor MQ' in smaller white font below it. The third box contains '5 det' in large light blue font and 'prediksi ke depan' in smaller white font below it. On the right side of the slide, there are decorative light blue concentric circular shapes.

Lab IoT - ITB

TCN

Gas LPG Early Warning

Temporal Convolutional Network untuk prediksi kebocoran gas 5 detik sebelum terjadi

3,234 data	8 sensor MQ	5 det prediksi ke depan
---------------	----------------	----------------------------

Masalah yang Dipecahkan



Tanpa AI

- Sensor hanya bisa deteksi saat gas SUDAH bocor
- Tidak ada peringatan dini
- Risiko terlambat merespons



Dengan TCN

- Prediksi gas bocor 5 detik sebelumnya
- Peringatan lebih awal = waktu respons lebih banyak
- Model ringan — bisa jalan di ESP32-S3

Task: Klasifikasi clean_air / lpg_detected 5 detik ke depan

Data & Sensor

MQ135

CO₂/NH₅

MQ2

LPG/Gas

MQ3

Alkohol

MQ4

CH₄

MQ7

CO

MQ5

LPG/Propana

MQ6

LPG/Butana

MQ8

H₂

Pembagian Data (urutan waktu)

Train 70%

Val
15%

Test
15%

3,234

total baris data

2

label: clean_air/lpg_detected

1 Hz

samplingrate sensor

Cara Model Membaca Data



Input: 15 detik lalu

Prediksi: detik ke-21

15 detik riwayat

Model melihat data 8 sensor selama 15 detik terakhir

+5 detik ke depan

Model memprediksi kondisi gas 5 detik setelah window berakhir

Tidak lintas sesi

Window tidak boleh menyeberangi sesi pengukuran yang berbeda

Arsitektur TCN



Cara kerja Dilated Convolution:

Dilasi 1 — melihat 3 titik berurutan



Dilasi 2 — melompat 1 titik, jangkauan 2x

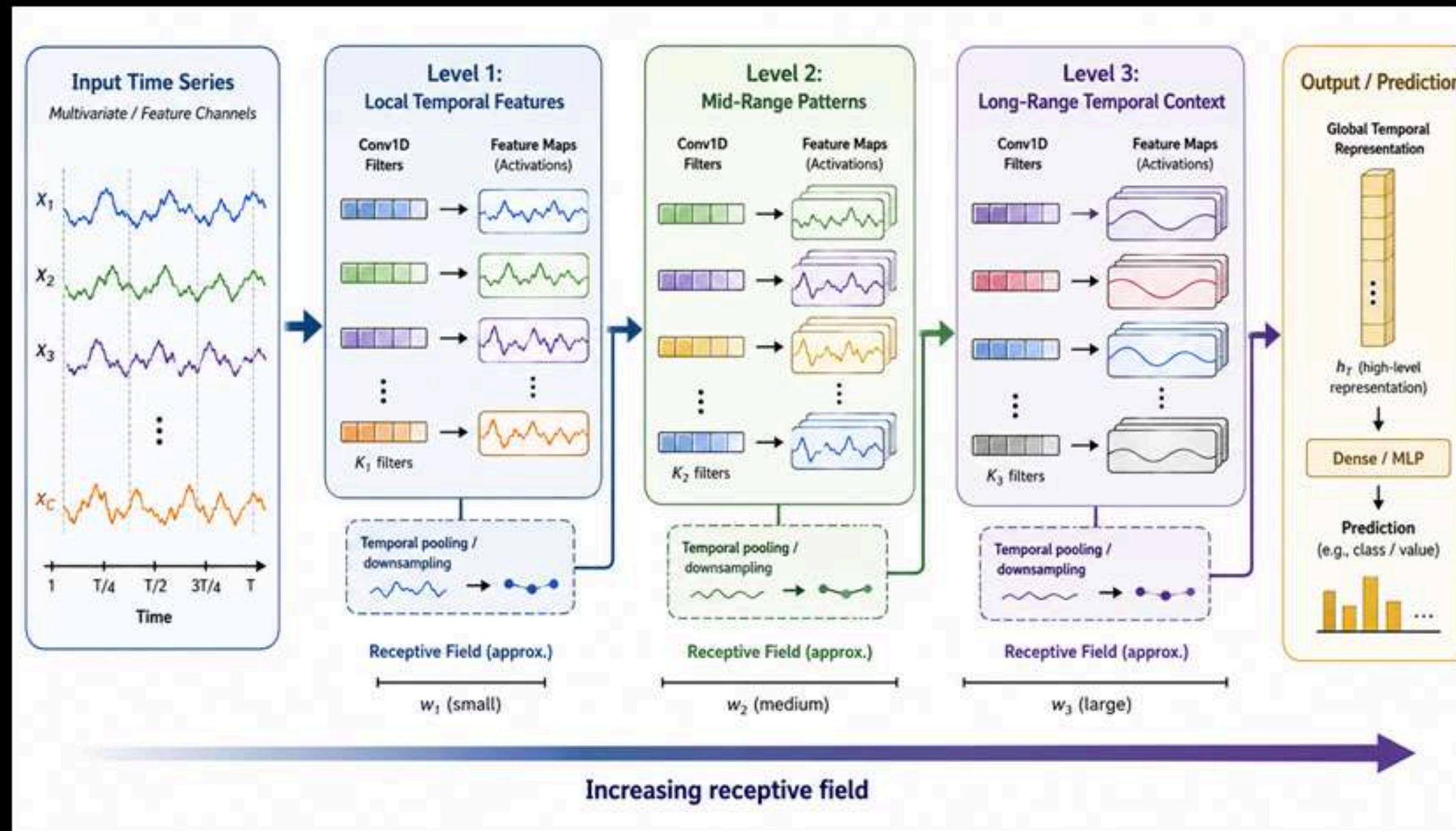


Dilasi 4 — melompat 3 titik, jangkauan 4x

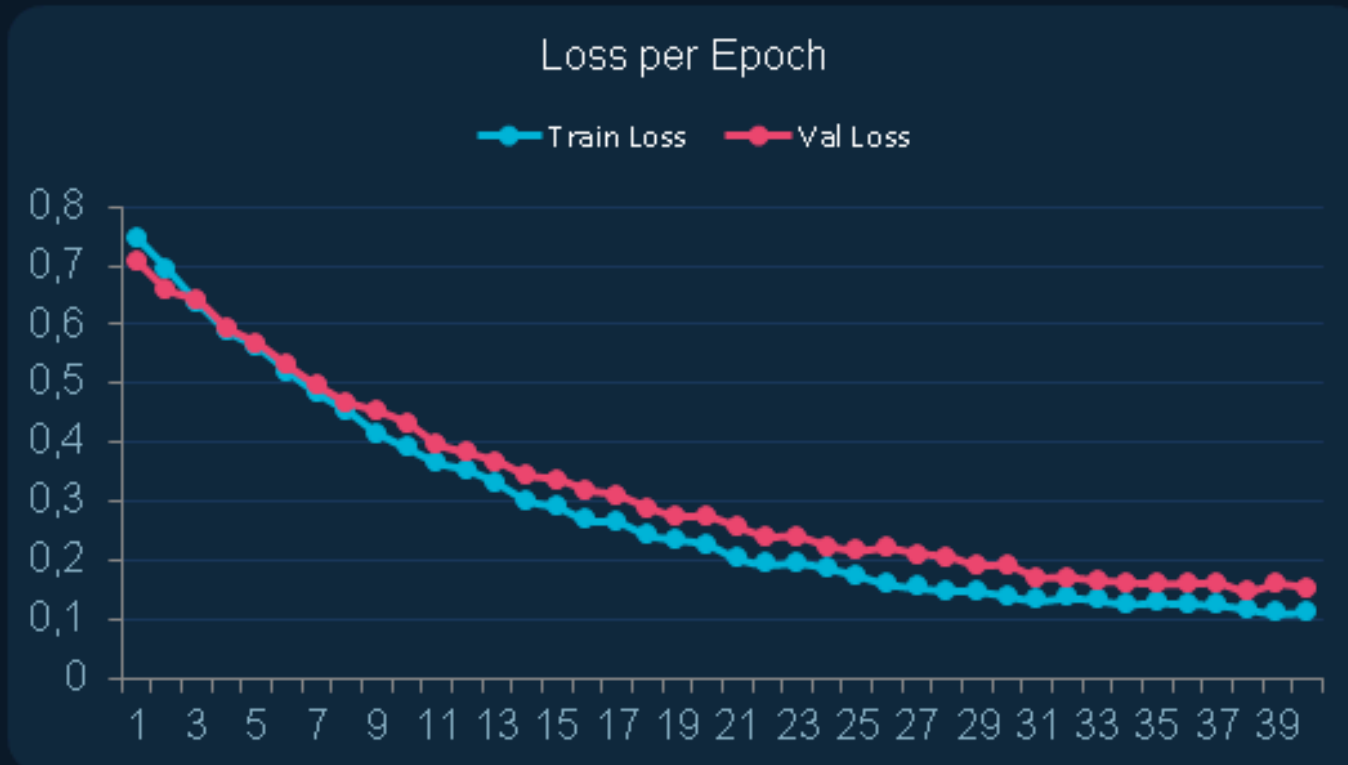


Receptive field: 31 timestep · Lookback: 15 timestep ☒ mencakup semua input · Total parameter: ~13,000

Arsitektur TCN



Proses Training



Class Weight

Kompensasi data tidak seimbang: jumlah clean_air ≠ lpg_detected

Early Stopping

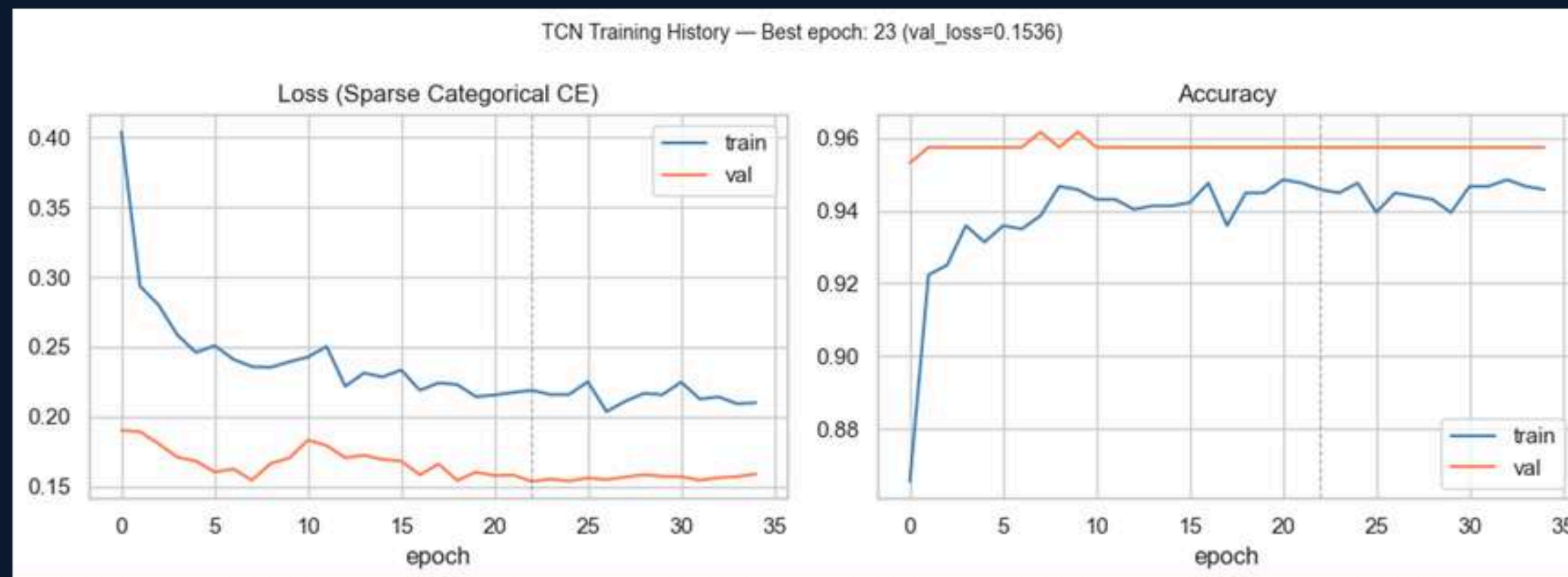
Hentikan training otomatis jika val_loss tidak membaik (patience 12)

Reduce LR

Kurangi learning rate 0.5× saat stagnan (patience 5)

Batch size: 32 · Max epoch: 100 · Optimizer: Adam (lr=0.001)

Hasil Training — Data Aktual



23

BestEpoch

0.154

Val Loss

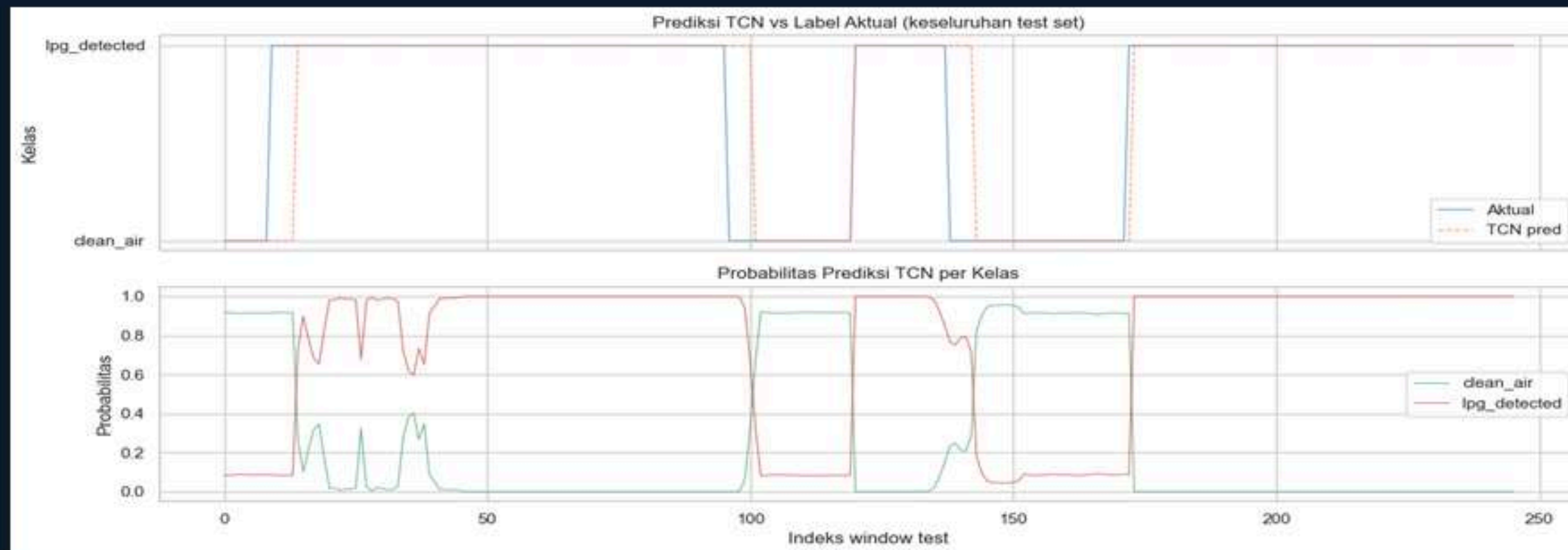
~96%

Val Accuracy

93.5%

Test Accuracy

Prediksi TCN vs Label Aktual



☑ Prediksi (garis putus-putus) hampir selalu mengikuti label aktual (garis solid)

⚡ Transisi `clean_air` → `lpg_detected` terdeteksi dengan sedikit delay — wajar untuk forecasting 5 detik ke depan

📊 Probabilitas prediksi mendekati 1.0 saat model sangat yakin, turun saat transisi

Prediksi TCN vs Label Aktual

	Index	Label Aktual	Label Prediksi	Prob Clean Air	Prob Gas Leak	Status
0	0	clean_air	clean_air	0.924909	0.075091	Benar
1	1	clean_air	clean_air	0.926923	0.073077	Benar
2	2	clean_air	clean_air	0.925294	0.074706	Benar
3	3	clean_air	clean_air	0.925324	0.074676	Benar
4	4	clean_air	clean_air	0.925870	0.074130	Benar
5	5	clean_air	clean_air	0.926445	0.073555	Benar
6	6	clean_air	clean_air	0.925910	0.074090	Benar
7	7	clean_air	clean_air	0.925059	0.074941	Benar
8	8	clean_air	clean_air	0.926681	0.073319	Benar
9	9	lpg_detected	clean_air	0.926271	0.073729	Salah
10	10	lpg_detected	clean_air	0.928506	0.071494	Salah
11	11	lpg_detected	clean_air	0.926714	0.073286	Salah
12	12	lpg_detected	clean_air	0.925236	0.074764	Salah
13	13	lpg_detected	clean_air	0.927091	0.072909	Salah
14	14	lpg_detected	lpg_detected	0.439581	0.560419	Benar
15	15	lpg_detected	lpg_detected	0.162870	0.837130	Benar
16	16	lpg_detected	lpg_detected	0.195288	0.804712	Benar
17	17	lpg_detected	lpg_detected	0.318442	0.681558	Benar
18	18	lpg_detected	lpg_detected	0.346168	0.653832	Benar
19	19	lpg_detected	lpg_detected	0.193250	0.806750	Benar

Hasil Test

TCN

93.5%

Test Accuracy

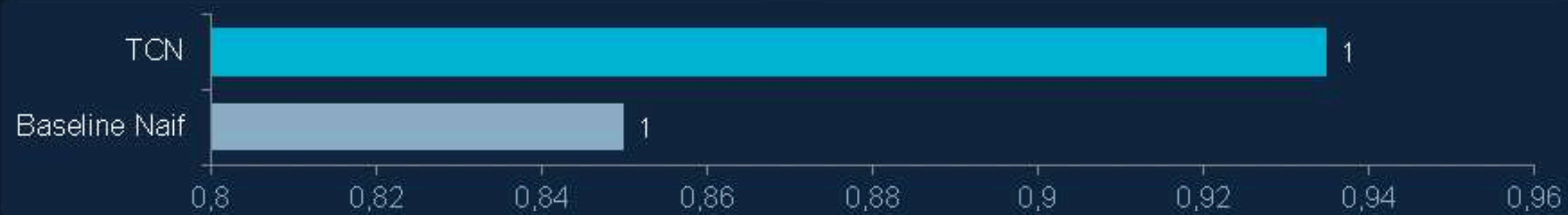
F1 Macro 0.916 · Loss 0.2562

Baseline Naif

~85%

(tebak label = sekarang)

F1 Macro ~ 0.82



Confusion Matrix — TCN vs Baseline

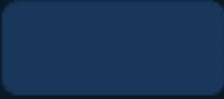
TCN (Akurasi 93.5%)

		Prediksi →	
		clean_air	lpg_detect
Aktual ↓	clean_air	57	10
	lpg_detect	6	173

Baseline Naif (~85%)

		Prediksi →	
		clean_air	lpg_detect
Aktual ↓	clean_air	52	15
	lpg_detect	22	157

 Prediksi benar

 Prediksi salah

TCN lebih sedikit salah prediksi di kedua kelas

Deployment ke ESP32-S3



Ukuran Model

Float32



INT8



13 KB

Kenapa INT8?

- 4x lebih kecil dari float32
- Lebih cepat di hardware MCU
- Akurasi hampir setara

Model siap deploy — ukuran INT8 ~13 KB, cocok untuk flash ESP32-S3

Kesimpulan



TCN berhasil prediksi kebocoran LPG 5 detik ke depan



Test accuracy 93.5%, F1 macro 0.916, loss 0.2562 — hasil nyata dari notebook



Training lebih cepat karena komputasi paralel



Model ringan: INT8 hanya ~13 KB — siap masuk ESP32-S3

TCN PER-SENSOR

Prediksi Kebocoran Gas LPG 5 Detik ke Depan

8 Model TCN Independen — Satu Model per Sensor MQ (Board12)

MQ135V

MQ2V

MQ3V

MQ4V

MQ7V

MQ5V

MQ6V

MQ8V

Dibuat dari notebook: Board12_TCN_PerSensor_8Model.ipynb

LATAR BELAKANG

Mengapa 8 Model Terpisah?

Board12 memiliki 8 sensor gas MQ. Notebook sebelumnya melatih 1 model TCN multivariat (menerima 8 fitur sekaligus). Notebook ini melatih 8 model TCN univariat — masing-masing hanya menerima 1 sensor sebagai input — untuk mengukur seberapa informatif tiap sensor secara individual dalam memprediksi kebocoran LPG 5 detik ke depan.

8

Model TCN
(1 per sensor)

2

Kelas target
clean_air / lpg_detected

5s

Horizon prediksi
ke depan

SETIAP MODEL

- Arsitektur TCN identik dengan notebook multi-sensor
- Dilatih, dievaluasi & diekspor secara independen
- Output: TFLite float32 + int8 (untuk edge device)
- Hasil dirangkum dalam tabel perbandingan 8 model

DATASET

Data Sensor Board12

3,234

total baris data
(seluruh sesi)

8

sensor gas MQ digunakan sebagai fitur

8 Sensor MQ

MQ135V

MQ2V

MQ3V

MQ4V

MQ7V

MQ5V

MQ6V

MQ8V

Label Klasifikasi (Biner)

0

clean_air

Kondisi udara bersih / normal

1

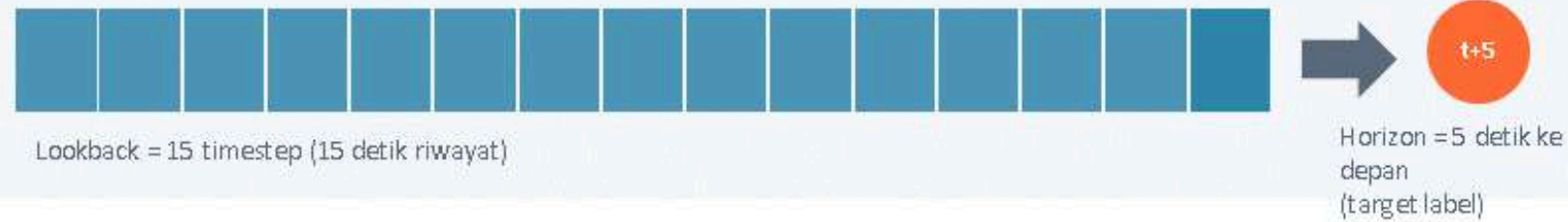
lpg_detected

Terdeteksi indikasi kebocoran LPG

PREPROCESSING

Windowing, Split & Normalisasi

Skema Windowing (per sesi, 1 Hz)



Split Temporal per Sesi

Train 70% / Val 15% / Test 15%, dibagi berurutan (bukan acak) di dalam tiap sesi rekaman.

Normalisasi Z-score

Mean & std dihitung hanya dari data train, lalu diterapkan ke val & test (mencegah data leakage).

Downsampling Sesi Monoton

Sesi dengan label dominan/konstan di-cap maksimum 350 baris agar data tidak timpang.

Temporal Convolutional Network (TCN)



Setiap TCN Block: $2 \times (\text{ZeroPad} \rightarrow \text{Conv1D dilated} \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{SpatialDropout}) + \text{koneksi residual (skip connection)}$

Filters

32

Kernel size

3

Dilation rates

1, 2, 4, 8

Dropout

0.2

Skip connection

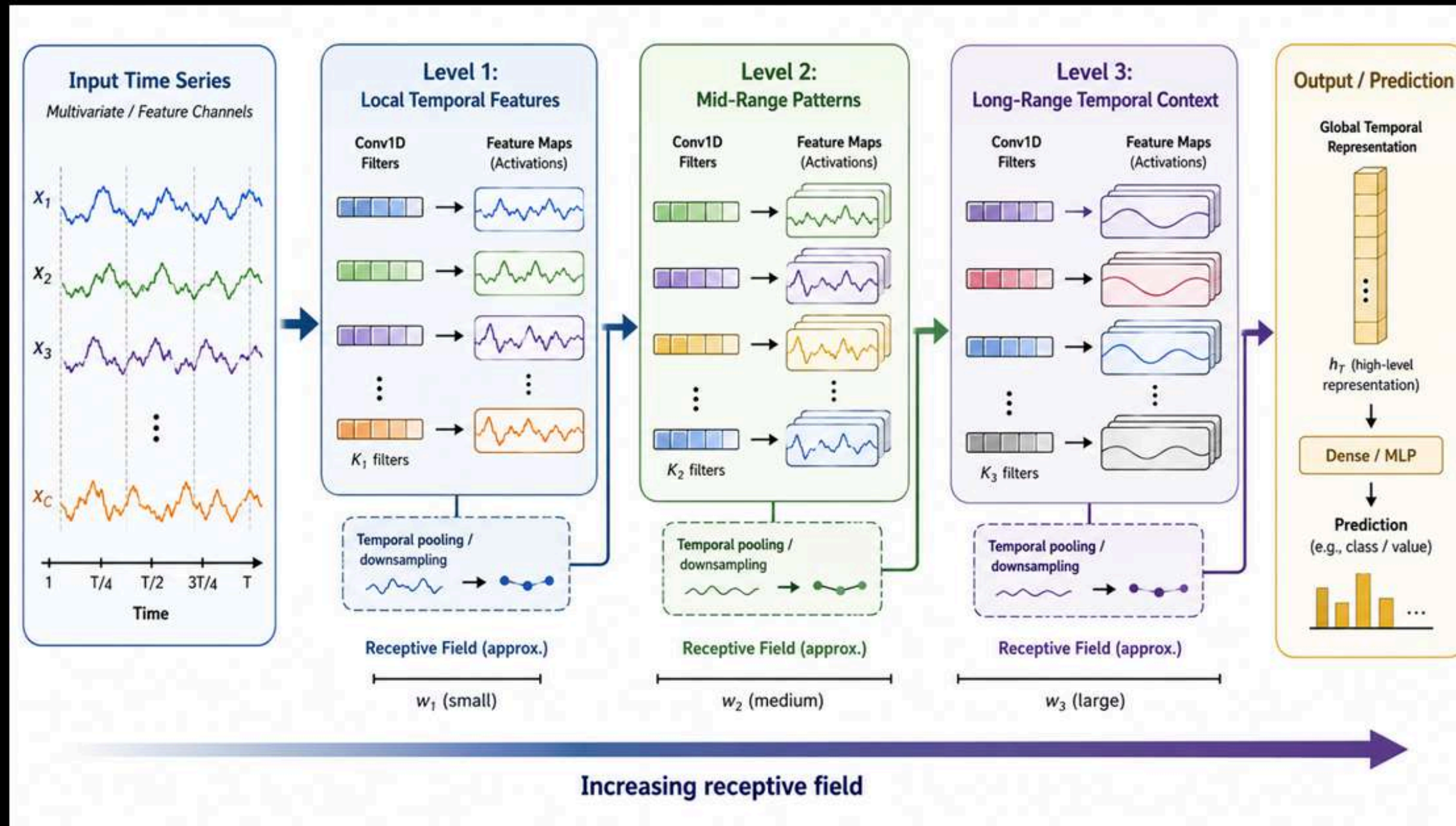
Ya

Receptive field

25 timestep ($\geq$ lookback 15 ✓)

Total parameter per model: 25,714 · Ukuran float32: ~131.8 KB

Arsitektur TCN



TRAINING

Strategi Pelatihan 8 Model

Loop untuk Setiap Sensor

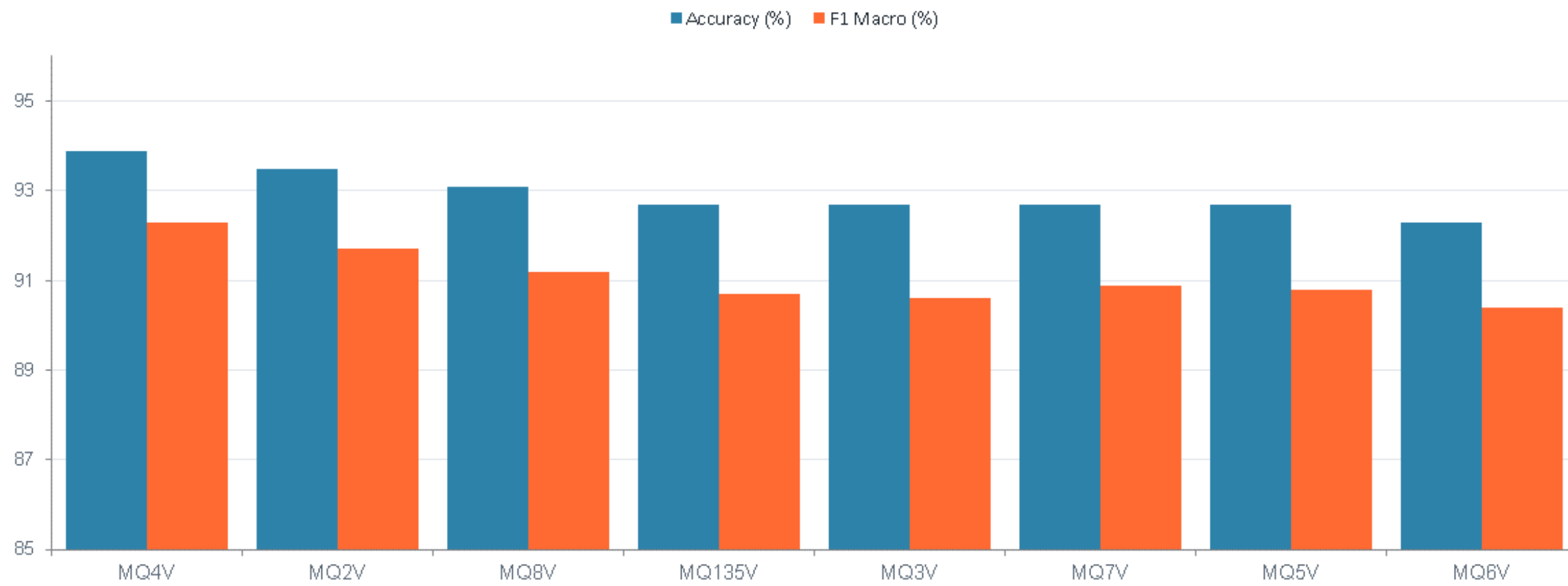
- 1 Bangun dataset windowed 1-fitur untuk sensor tsb.
- 2 Bangun model TCN baru (`n_features = 1`)
- 3 Hitung class weight (balanced) untuk atasi imbalance
- 4 Latih dengan EarlyStopping + ReduceLROnPlateau + ModelCheckpoint
- 5 Evaluasi di test set (accuracy, F1, confusion matrix)
- 6 Ekspor ke TFLite float32 & int8 + simpan norm params

KONFIGURASI TRAINING

Optimizer	Adam (lr = 1e-3)
Loss	Sparse categorical crossentropy
Batch size	32
Max epochs	100 (dengan EarlyStopping)
EarlyStopping	patience = 12, monitor val_loss
ReduceLROnPlateau	factor 0.5, patience 5, min_lr 1e-5
Class weight	balanced (mengatasi imbalance kelas)
Seed	42 (reproducible)

HASIL

Perbandingan Akurasi 8 Model



Sensor diurutkan dari performa tertinggi ke terendah — semua 8 model mencapai akurasi ≥ 92%

HASIL

Ringkasan Akhir 8 Model

Sensor	Accuracy	F1 Macro	Precision (lpg)	Recall (lpg)	Best Epoch	Float32 / Int8 (KB)
MQ4V	0.939	0.923	0.961	0.955	11	131.8 / 72.3
MQ2V	0.935	0.917	0.950	0.961	27	131.8 / 72.3
MQ8V	0.931	0.912	0.950	0.955	20	131.8 / 72.3
MQ135V	0.927	0.907	0.945	0.955	14	132.2 / 72.5
MQ3V	0.927	0.906	0.940	0.961	29	131.8 / 72.3
MQ7V	0.927	0.909	0.955	0.944	10	131.8 / 72.3
MQ5V	0.927	0.908	0.950	0.950	15	131.8 / 72.3
MQ6V	0.923	0.904	0.955	0.939	10	131.8 / 72.3

MQ4V menunjukkan performa tertinggi secara individual untuk mendeteksi kebocoran LPG 5 detik ke depan

DEPLOYMENT

Export Model untuk Edge Device

.keras

Keras Checkpoint

Bobot terbaik disimpan otomatis via ModelCheckpoint (monitor val_loss).

~131.8 KB

TFLite Float32

Model presisi penuh, cocok untuk validasi & referensi akurasi.

~72.3 KB

TFLite Int8

Model terkuantisasi (representative dataset), kompresi ~1.8x dari float32.

.json

Norm Params

Mean, std, lookback, horizon & label map — dibutuhkan saat inferensi di device.

Setiap sensor menghasilkan keempat artefak di atas → total 8 set artefak, disimpan di folder `tcn_per_sensor_outputs/`

KESIMPULAN

Setiap Sensor MQ Cukup Informatif Sendiri

- **Semua 8 model $\geq 92\%$ akurasi**

Bahkan dengan hanya 1 fitur, tiap model mampu memprediksi kebocoran LPG 5 detik ke depan dengan andal.

- **MQ4V adalah sensor paling informatif**

Akurasi 93.9%, F1 macro 92.3% — tertinggi di antara 8 sensor, diikuti MQ2V dan MQ8V.

- **Model ringan & siap edge deployment**

~25.7K parameter, ukuran int8 hanya ~72 KB — cocok untuk mikrokontroler di board sensor.

- **Arsitektur konsisten & dapat dibandingkan**

Karena hyperparameter & pipeline identik antar model, perbandingan antar sensor menjadi adil (apples-to-apples).

CLUSTER HEAD / LORA MESH NETWORK

PROGRESS LORA MESH NETWORK

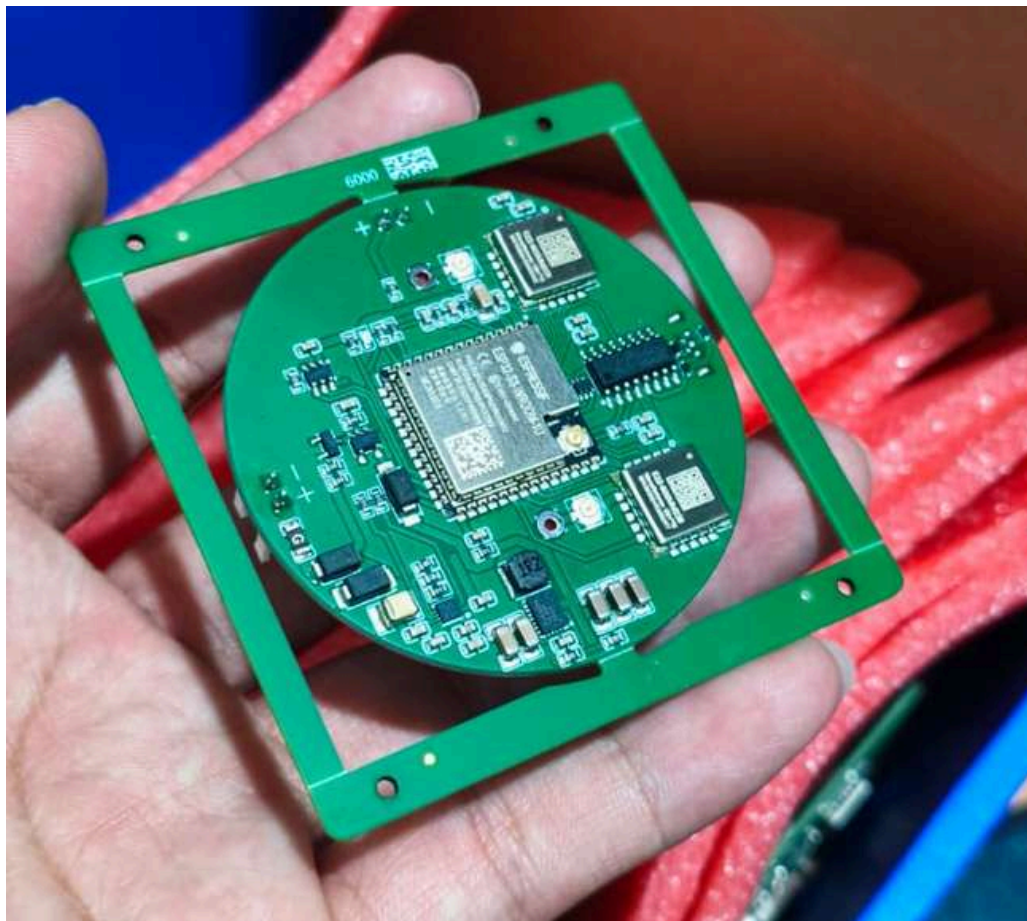


Cluster Head v1

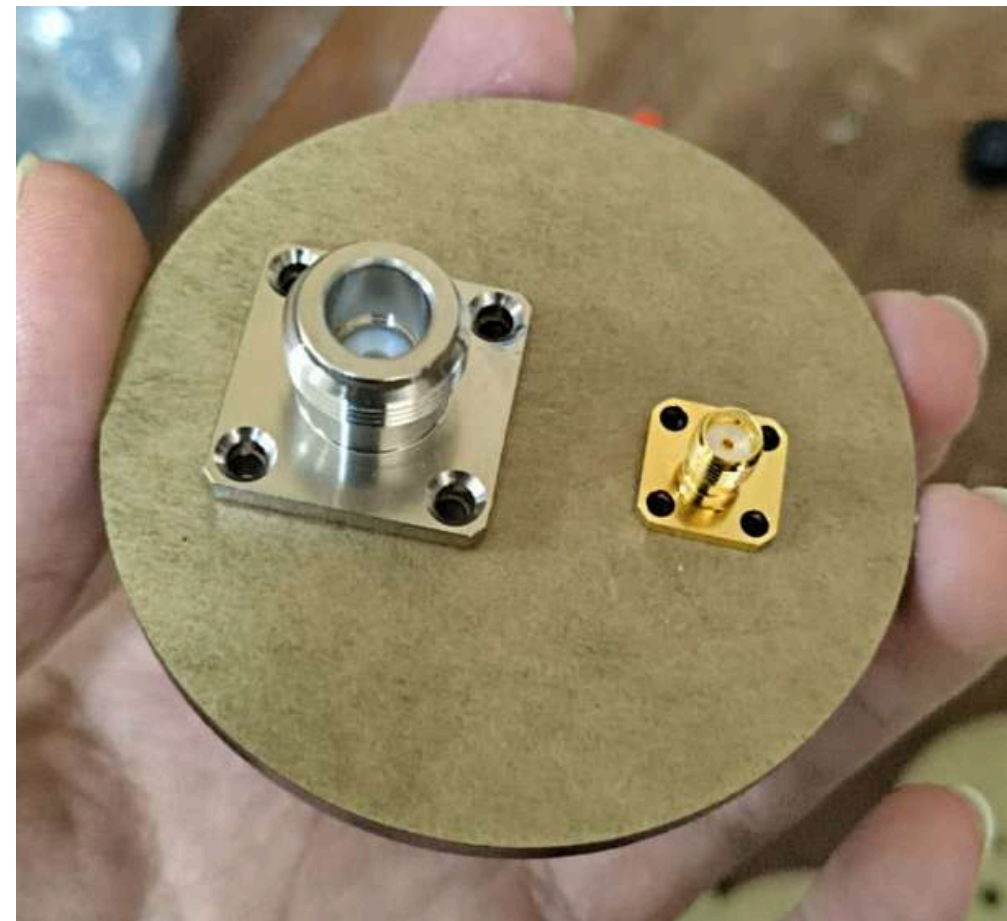


Cluster Head v2

PROGRESS LORA MESH NETWORK



Cluster Head v3



CH Case



Fiberglass Antenna

Desain CH v3 dikembangkan untuk mendukung penggunaan antenna fiberglass dengan gain dBi yang lebih tinggi. Penggunaan antenna ini diharapkan dapat memperluas cakupan jaringan, meningkatkan kualitas link komunikasi, serta mendukung performa jaringan LoRa yang lebih stabil pada skenario field testing.

Desain Tiang Cluster Head & Bracket Solar Panel



PROGRESS LORA MESH NETWORK

Pengujian Arsitektur Jaringan Mesh di Lab

Topologi Normal



Alur Inisialisasi

CH3/GW aktif pertama → CH2 config → CH1 config →
Node kirim data

Validasi Link

CH memilih parent berdasarkan respons dan RSSI terbaik;
ACK diterapkan pada setiap hop.

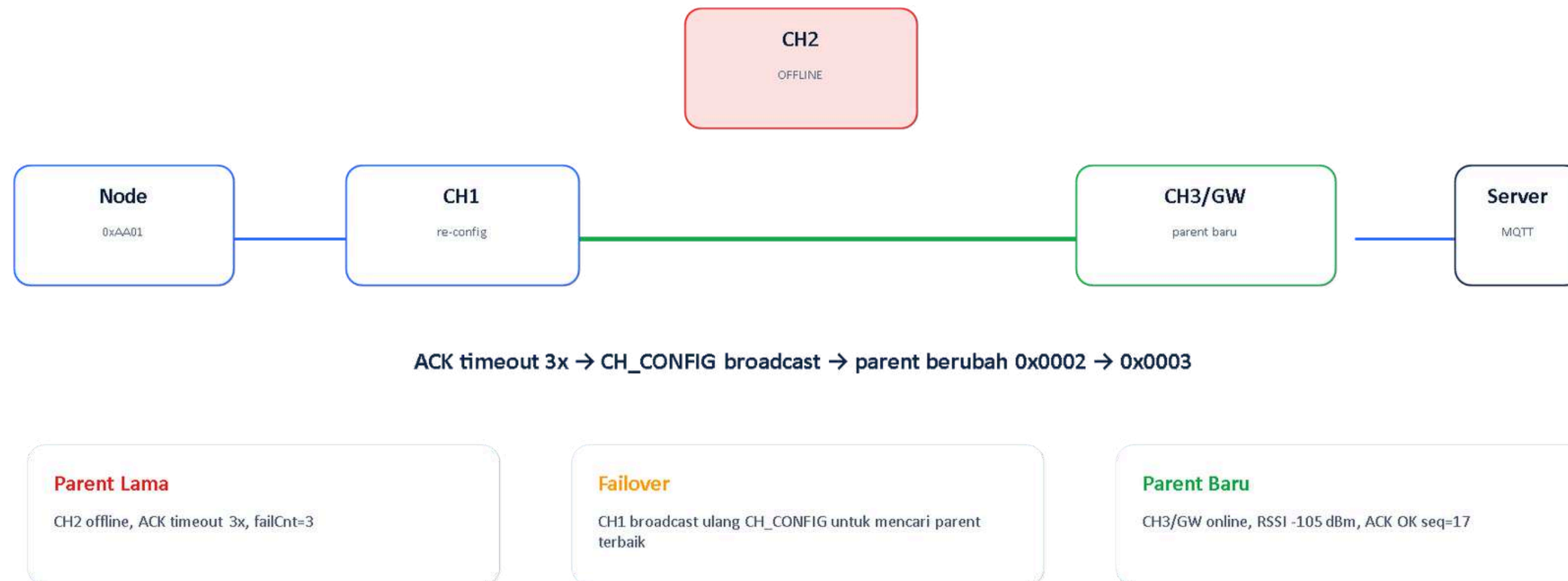
Hasil

Data flow Node → CH1 → CH2 → CH3/GW → Server
berhasil.

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Pengujian Arsitektur Jaringan Mesh di Lab

Topologi Failover



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Pengujian Arsitektur Jaringan Mesh di Lab

Kesimpulan

Konfigurasi Otomatis

Semua CH berhasil discovery dan memilih parent berdasarkan RSSI terbaik.

Failover Otomatis

Saat CH2 offline, CH1 pindah parent ke CH3/GW setelah timeout 3x.

Topologi Star-Mesh

Node → CH1 → CH2 → CH3/GW berjalan dengan ACK per hop.

Re-routing Transparan

Data tetap terkirim via jalur baru tanpa intervensi manual.

PROGRESS LORA MESH NETWORK

Pengujian Arsitektur Jaringan Mesh



Simulasi awal pengukuran LoRa menggunakan LCD sebagai media pengecekan data, sehingga tidak memerlukan laptop atau handphone.

Metode ini disiapkan untuk mengantisipasi kemungkinan pembatasan perangkat elektronik saat site survey di RU.

UJI JARAK DAN RELIABILITAS KOMUNIKASI KOMUNIKASI DATA

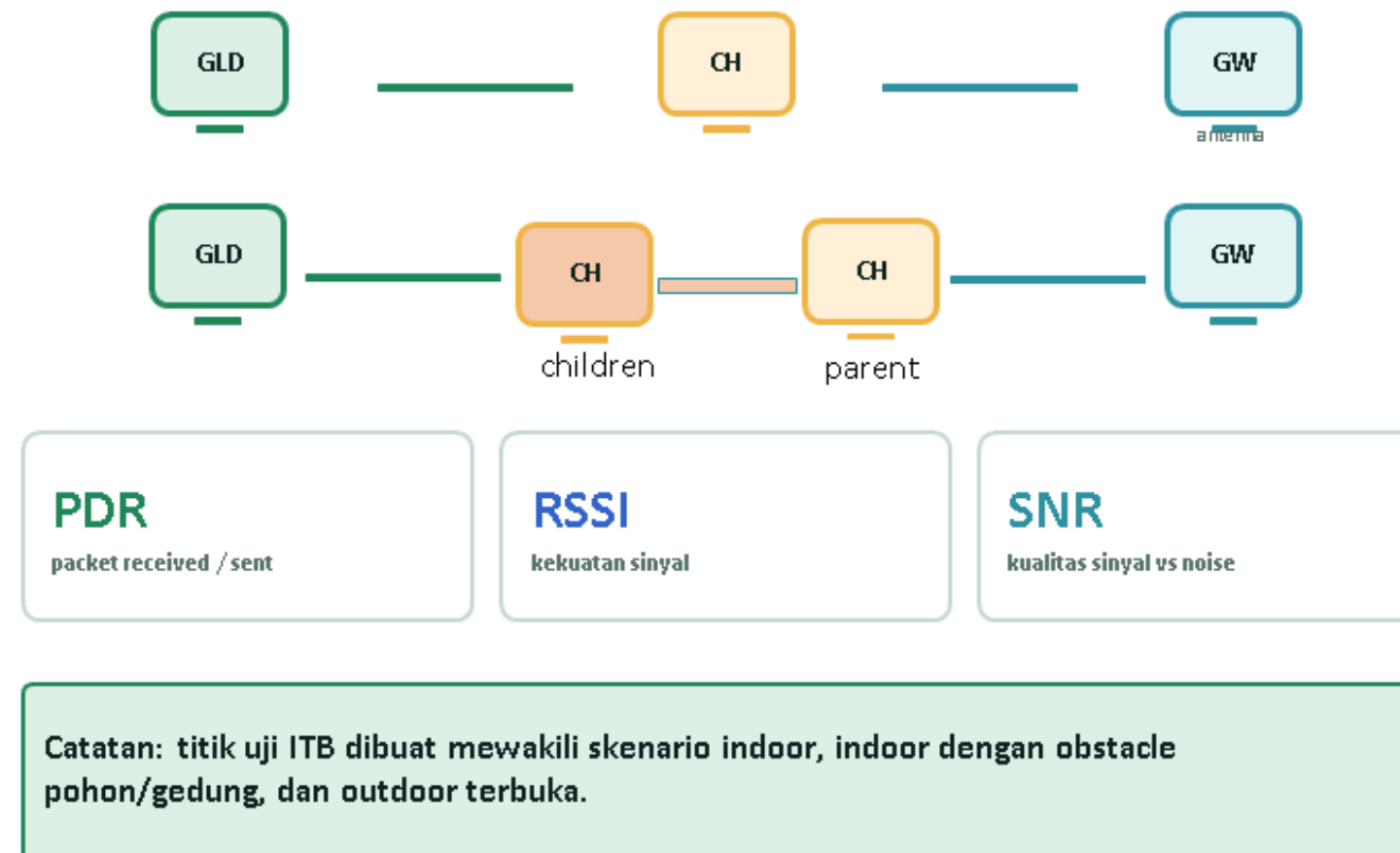
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Pengujian Arsitektur Jaringan Mesh

Topologi uji: GLD -> CH -> GW



Simulasi dilakukan sebelum site survey untuk memperoleh baseline performa sistem, meliputi jangkauan komunikasi, kualitas link, mekanisme failover, dan ketahanan Cluster Head (CH) pada kondisi operasional luar ruangan (outdoor).

1. Uji paket data dan jarak

GLD dummy beberapa titik. Catat jarak GLD-CH-GW, RSSI, SNR, PDR, latency, dan obstacle.

2. Uji failover CH

Matikan beberapa CH. Pastikan A-CH mencari parent baru dan data tetap sampai ke GW.

3. Uji daya CH outdoor

Nyalakan CH dengan solar/baterai, enclosure outdoor, dan logging minimal satu minggu.

Parameter yang dicatat per test point

- Koordinat TX/RX dan elevasi.
- LOS/NLOS, obstacle gedung/pohon/beton.
- Konfigurasi radio: SF, BW, TX power, antenna.
- Packet sent, received, PDR, RSSI, SNR, retry.

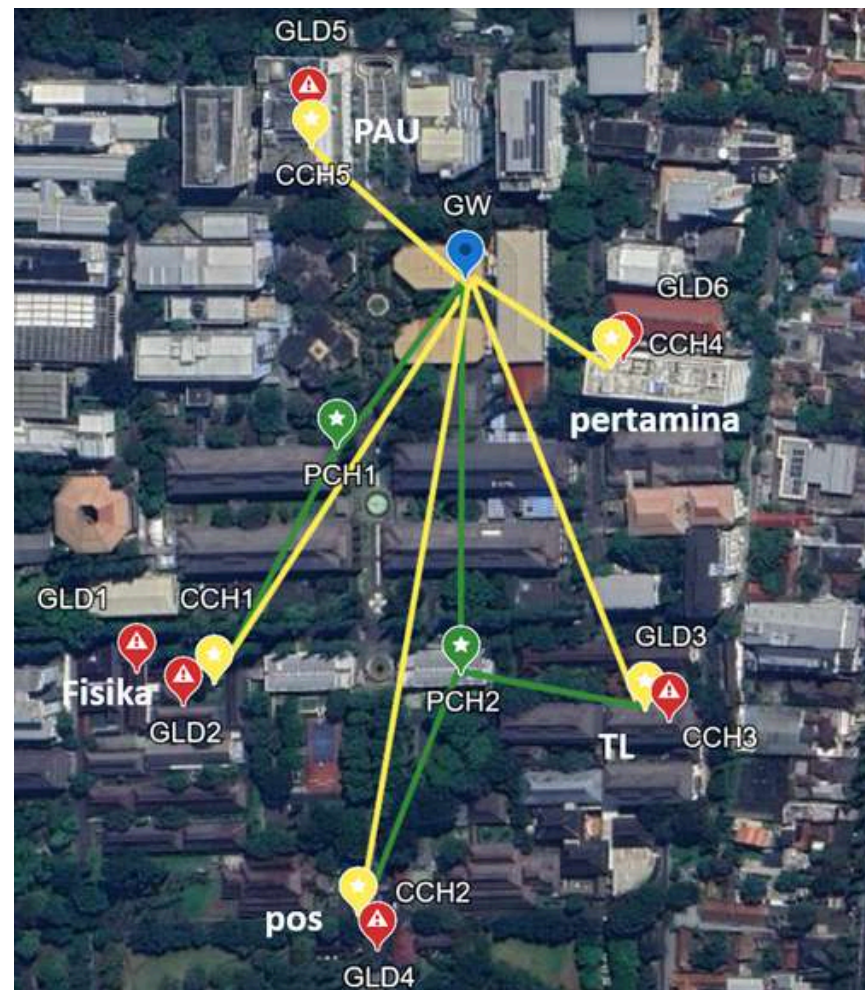
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Pengujian Arsitektur Jaringan Mesh

MAPPING AREA MENENTUKAN TITIK GLD DAN CH



GLD dalam ruangan area gedung

Prioritas di dekat sumber kebocoran: valve, flange, regulator, storage, pipa rawan vibrasi.

GLD dalam ruangan + banyak pohon

Perlu uji NLOS, multipath, dan posisi CH yang tidak tertutup kanopi/struktur.

GLD luar ruangan

Perhatikan enclosure, paparan hujan/panas, korosi, vandal risk, dan akses maintenance.

Logika penentuan CH

RSSI stabil

SNR aman

PDR 100%

CH final dipilih dari titik yang memberi margin RF terbaik, bukan hanya jarak terdekat. Jika PDR turun saat ada obstacle, kandidat CH harus digeser atau ditambah repeater/CH perantara.

Mapping area digunakan sebagai dasar penentuan titik pemasangan GLD dan Cluster Head. Titik GLD diprioritaskan pada area dengan potensi risiko kebocoran gas, seperti valve, flange, pipa, regulator, storage, atau peralatan proses. Sementara itu, titik CH ditentukan berdasarkan validasi kualitas RF, meliputi RSSI, SNR, PDR, obstacle, dan kondisi line-of-sight.

Cluster Head final dipilih dari lokasi yang memberikan margin komunikasi paling aman, bukan hanya dari jarak terdekat terhadap node. Jika terjadi penurunan PDR akibat obstacle atau kondisi NLOS, maka posisi kandidat CH perlu disesuaikan atau ditambahkan repeater/CH perantara untuk menjaga reliabilitas jaringan.

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MEKANISME PENGUJIAN



cek RSSI/SNR/PDR saat CH di 5-6 m

CH mengaktifkan WiFi AP lokal + halaman HTML dashboard saat mode survei.
Laptop/smartphone konek ke WiFi CH dari bawah, lalu membaca RSSI, SNR, PDR, packet counter, dan log test secara real-time.

Tidak perlu USB panjang

Real-time dari bawah

Cocok untuk survei

Catatan deployment: WiFi AP debug hanya aktif pada mode survei/maintenance, memakai password, timeout otomatis, dan dimatikan saat operasi normal.

Tujuan pengujian adalah menentukan jarak maksimum yang masih mampu mempertahankan Packet Delivery Ratio (PDR) sebesar 100% dengan nilai RSSI dan SNR yang berada dalam batas margin aman, serta mendokumentasikan kondisi obstacle dan koordinat lokasi pengujian.

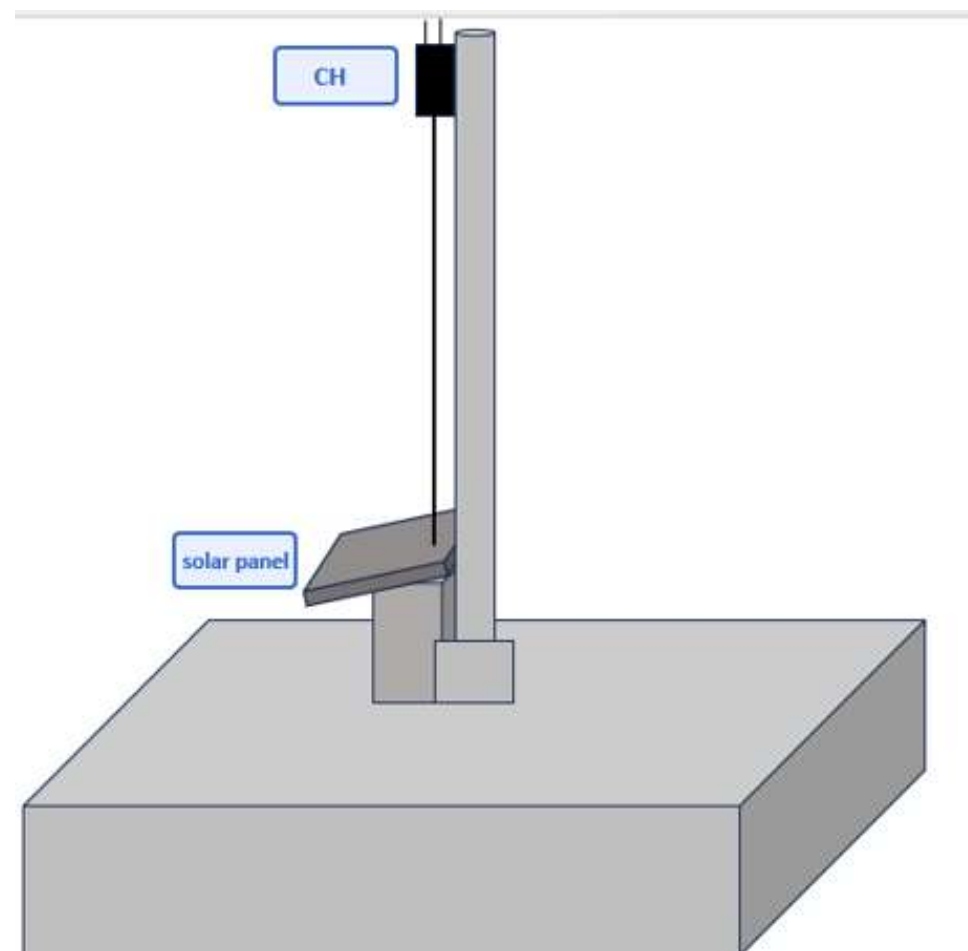
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Pengujian Arsitektur Jaringan Mesh

PELETAKAN NODE CH DI ROOFTOP



Rekomendasi dudukan

Diskusi dengan pak Totong untuk simulasi di ITB, di lapangan RU disesuaikan kembali

Stabilitas tiang 6 m

Periksa beban angin dan batas rooftop.

Penempatan panel surya

Panel kecil di bracket rendah pada base/ballast, bukan di puncak tiang, agar momen angin tidak bertambah besar.

Penempatan CH

CH IP65/IP66 di ujung tiang atau sedikit di bawah ujung tiang

Catatan HSE/struktur

Desain final harus dikonfirmasi oleh PIC fasilitas/struktur. Cek akses rooftop, proteksi petir/grounding, jalur kabel, dan izin kerja di ketinggian.

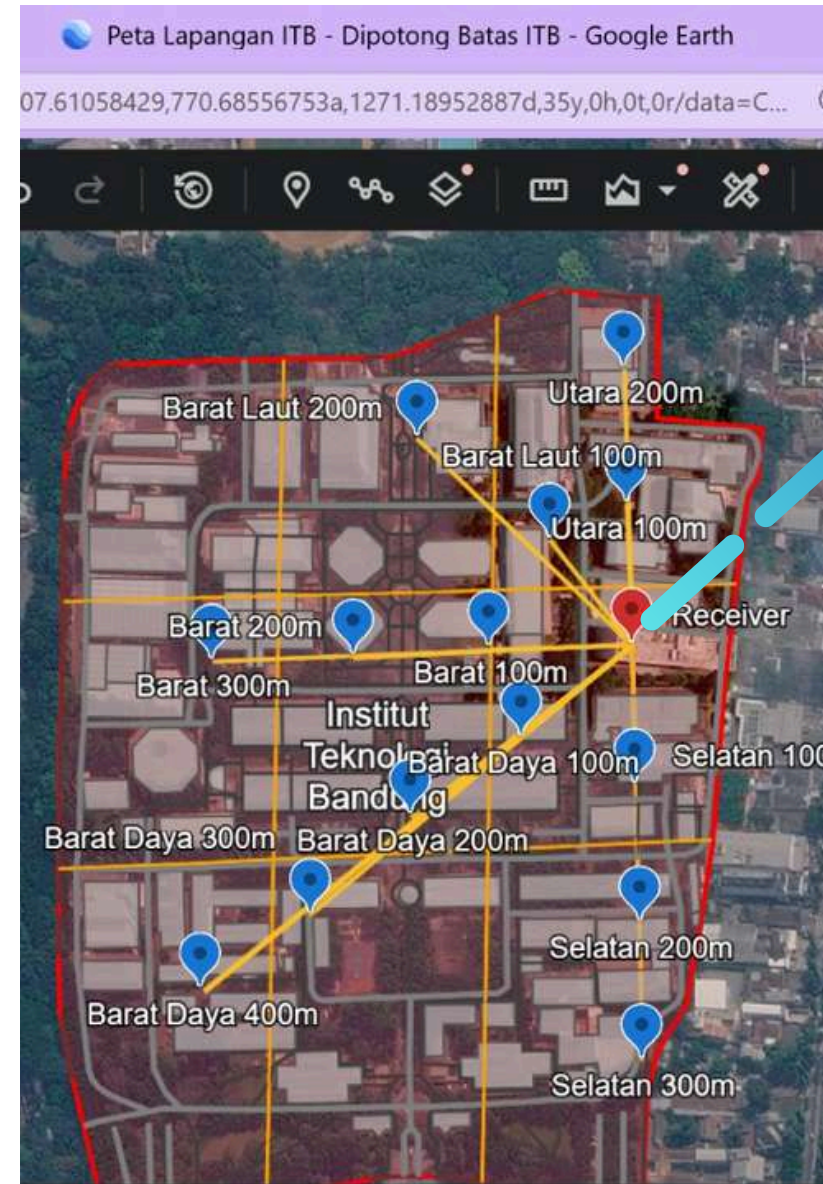


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Pengujian Arsitektur Jaringan Mesh



Pengujian jaringan **LoRa Mesh** dilakukan dengan menempatkan satu **Cluster Head (CH)** pada salah satu gedung tertinggi di kawasan kampus ITB untuk memperoleh cakupan komunikasi yang optimal serta mengevaluasi kualitas link pada beberapa titik pengujian.

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Pengujian Arsitektur Jaringan Mesh

OUTPUT PENGUJIAN AGAR SISTEM LAYAK DIBAWA KE SITE SURVEY

Kriteria utama layak dibawa ke RU

- PDR target 100% pada jarak kerja yang dipilih.
- RSSI/SNR tidak berada di batas minimum; harus punya margin.
- Failover terbukti, bukan hanya diasumsikan dari desain.
- Power CH outdoor punya rencana guard, logging, dan proteksi.

Sebelum pelaksanaan site survey di RU, simulasi harus menghasilkan bukti teknis yang dapat digunakan sebagai dasar penyusunan **mapping, Detail Engineering Design (DED)**, dan perencanaan pelaksanaan instalasi.



RF baseline per skenario area

Minimal indoor, indoor + obstacle, dan outdoor. Ada tabel jarak, RSSI, SNR, PDR, dan konfigurasi radio.



Topologi GLD-CH-GW awal

Ada candidate parent, rute failover, dan keputusan perlu CH tambahan/repeater.



Bukti failover

Saat CH dimatikan, node/CH lain menemukan parent baru dan data sampai ke GW.



Bukti daya outdoor

CH menyala stabil dengan solar/baterai/enclosure minimal 1 minggu atau punya rencana uji lanjut.



Dokumentasi lapangan siap

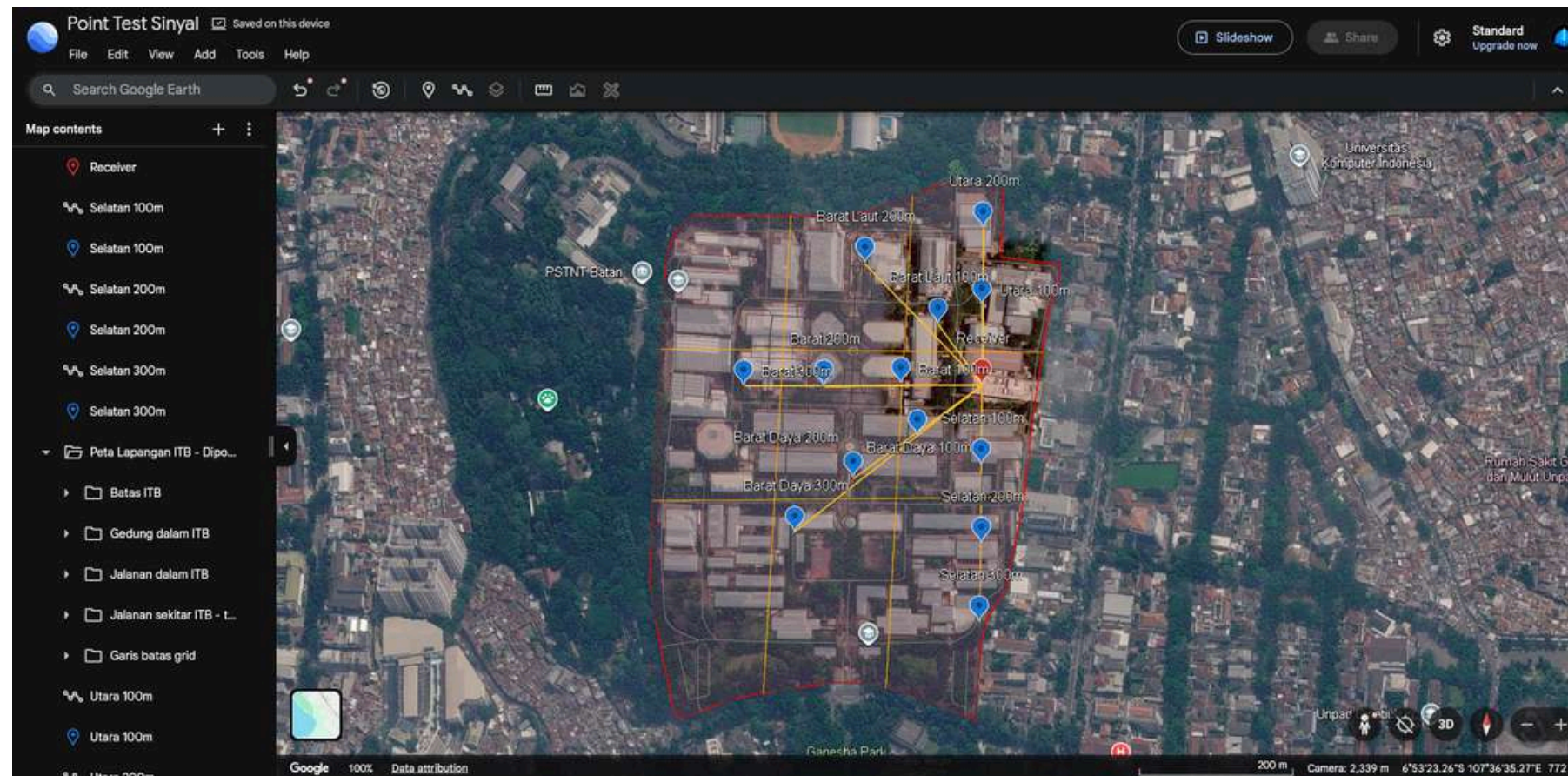
Template log, foto, koordinat, checklist, dan acceptance threshold siap dibawa ke RU.

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Pengujian Arsitektur Jaringan Mesh



Hasil pengujian awal menunjukkan bahwa cakupan satu Cluster Head (CH) masih terbatas dan belum mampu menjangkau seluruh titik pengujian. Untuk meningkatkan coverage jaringan LoRa Mesh, diperlukan penambahan Cluster Head pada lokasi dengan elevasi yang lebih tinggi serta jalur komunikasi line-of-sight yang lebih optimal.

Hasil pengujian ini menjadi acuan awal dalam menentukan strategi penempatan Cluster Head pada tahap site survey, penyusunan Detail Engineering Design (DED), dan pelaksanaan instalasi di lapangan.

PROGRESS LORA MESH NETWORK

Bagian Teknis yang perlu dipastikan saat survey



Metode kerja survei yang disarankan

Walkdown area bersama PICRU → kodekan setiap kandidat titik → uji LoRa pada titik kritis → validasi power/server

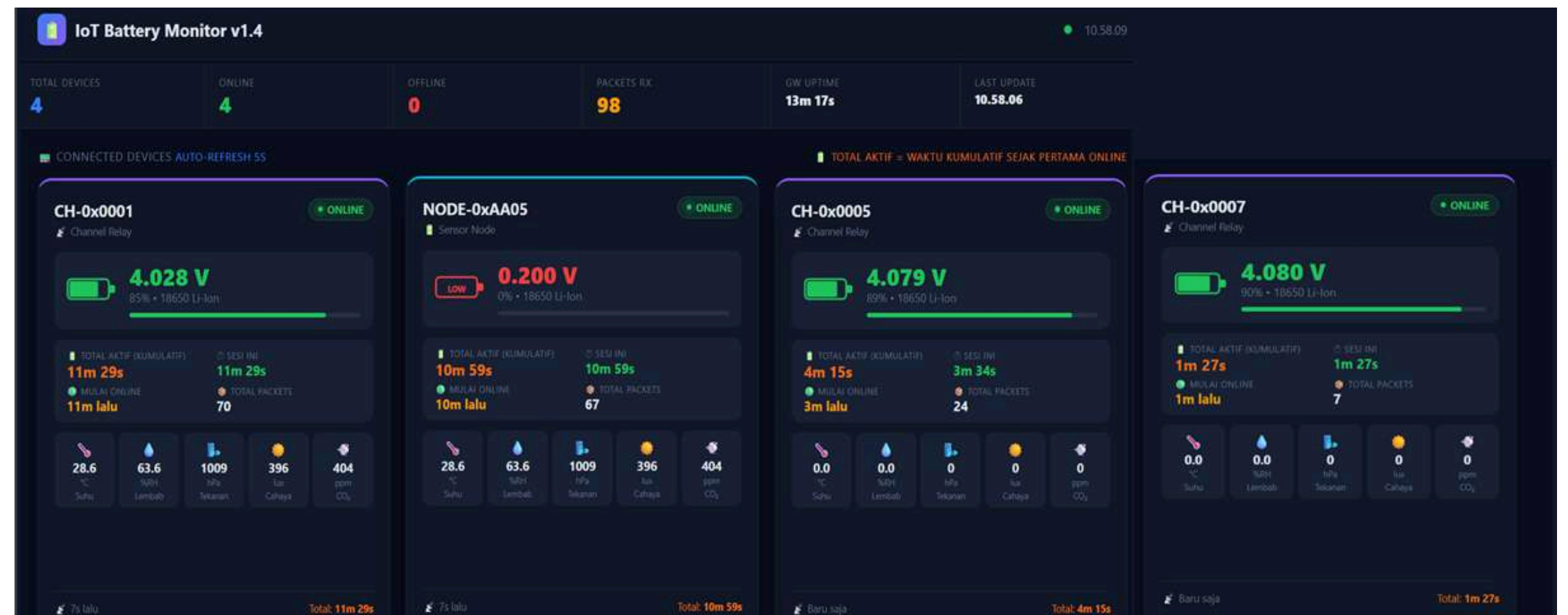
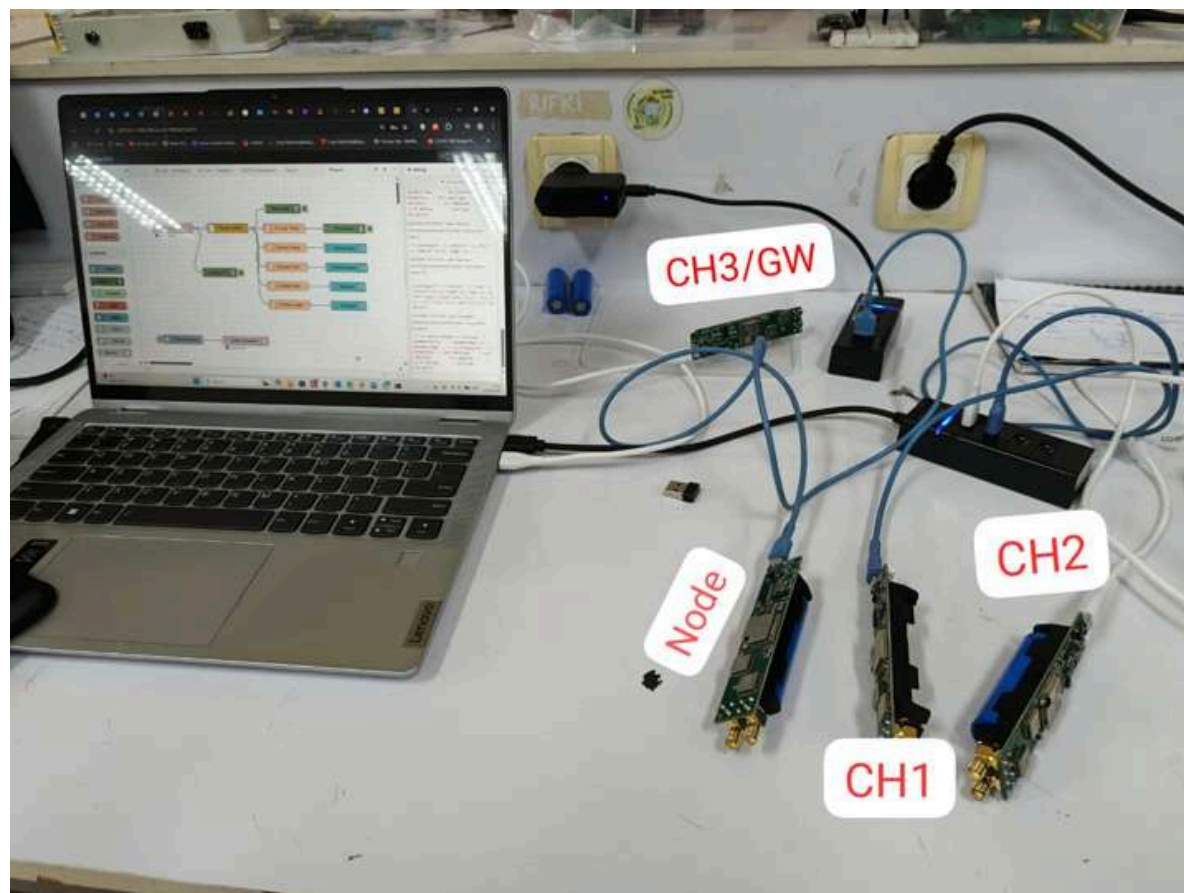


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PROGRESS LORA MESH NETWORK

Uji Konsumsi Panel Surya dan Baterai CH

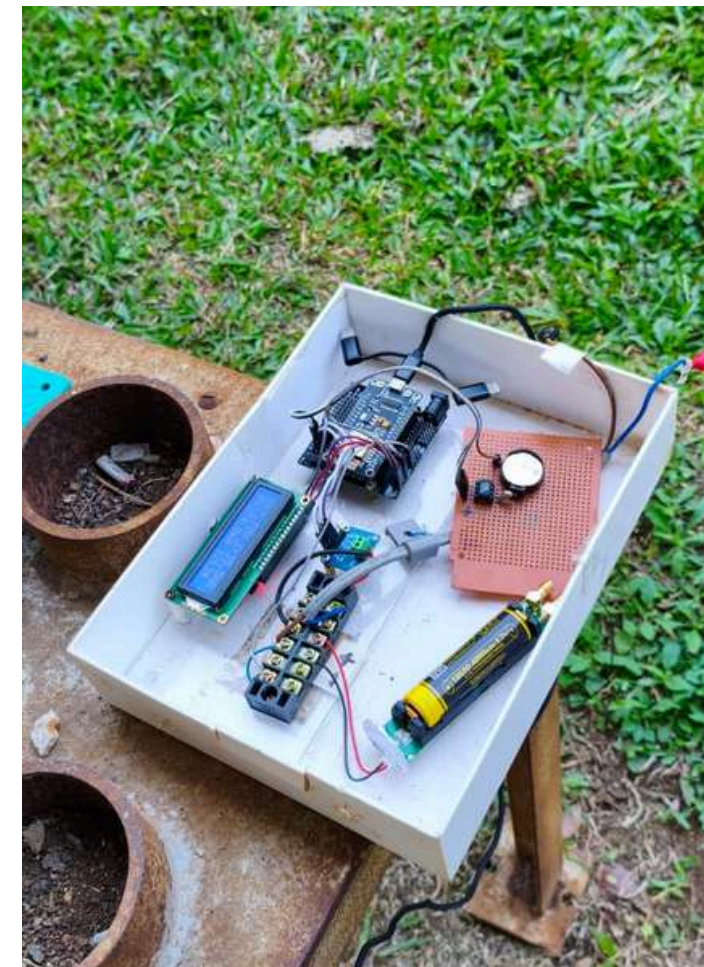
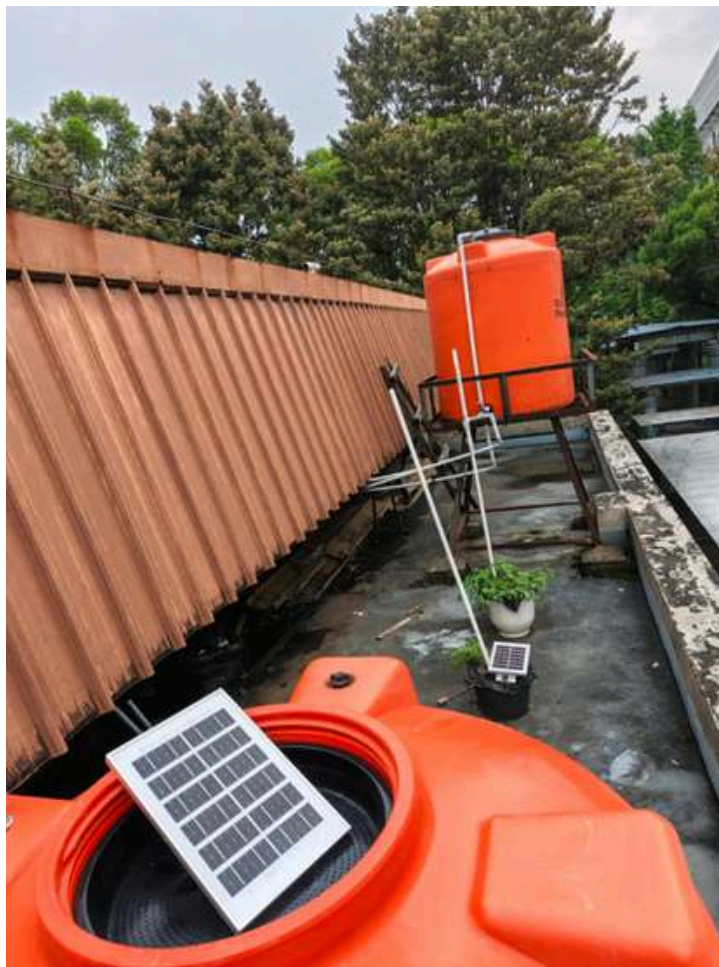


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PROGRESS LORA MESH NETWORK

Uji Konsumsi Panel Surya dan Baterai CH



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PROGRESS LORA MESH NETWORK

Uji Konsumsi Panel Surya dan Baterai CH



Baterai yang digunakan telah ditingkatkan menggunakan **LiitoKala**. Berdasarkan hasil pengujian, penggunaan satu baterai **LiitoKala** menunjukkan efisiensi konsumsi daya yang lebih baik dibandingkan konfigurasi tiga baterai **18650** konvensional.

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PROGRESS LORA MESH NETWORK

Uji Konsumsi Panel Surya dan Baterai CH

KESIMPULAN JUNI 2026

Berdasarkan hasil pengujian, panel surya telah berfungsi dengan baik dalam mengisi baterai sehingga kebutuhan daya **Cluster Head (CH)** tetap terpenuhi dan perangkat dapat beroperasi secara normal.

Proses optimasi masih dilakukan melalui evaluasi hambatan pada rangkaian, kualitas kabel, rangkaian pengisian (***charging circuit***), serta kesesuaian kapasitas daya (***wattage***) panel surya untuk meningkatkan efisiensi dan keandalan sistem.

SERVER APPLICATION



Single RU Dashboard

GASGUARD v2.1 — Per-Refinery-Unit Monitoring Portal
SCOPED DEPLOYMENT • ONE INSTANCE PER RU • RU2 + RU7

Next.js 15

GraphQL + Prisma

AI Classification

LoRa Mesh

Mapbox GL

SQLite

9

Screens

RU3

Demo Unit

9

AI Gas Classes

24/7

Per-RU Monitor



1 / 12



MOCK-UP DASHBOARD MONITORING

Secure Entry Point

Email Authentication



RU-Scoped Session



Restricted Access Banner



Production Plan

Next.js 15

GraphQL login

LocalStorage session

NestJS auth



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MOCK-UP DASHBOARD MONITORING

RU Overview Dashboard

Real-time command centre for one Refinery Unit. KPI cards, per-sensor gas heatmap, and device-type health breakdown — all scoped to the active RU.

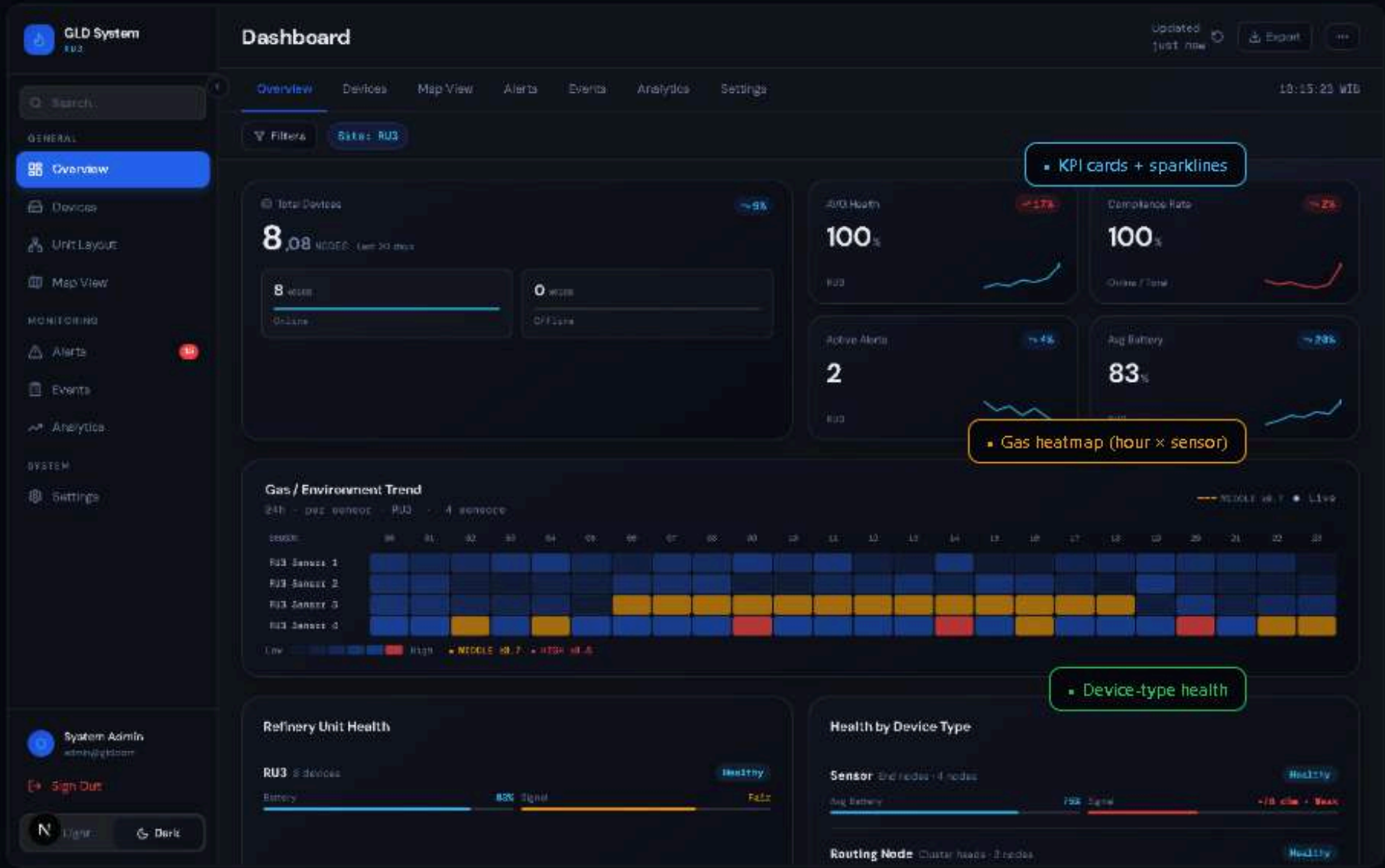
- ### KPI Cards

Total devices, online count, avg health %, compliance rate, active alerts, avg battery — each with sparkline trend.
- ### Gas / Environment Heatmap

Hour × sensor grid (00–23h). Cell color = AI confidence level: dark blue = low/none, amber = MIDDLE risk, red = HIGH risk.
- ### Refinery Unit Health

Battery % and signal strength (RSSI dBm) per RU row — instantly spot which layer needs maintenance.
- ### Health by Device Type

Sensor / Cluster Head / Gateway rows with avg battery and signal bars — pinpoint which tier is degrading.



Device Fleet Table

Full Fleet Table

Type Filter + Sort

Inline Rename

Alert Highlight

GraphQL devices query

Prisma

SQLite

```
updateDeviceName mutation
```


Network Topology

Visual LoRa mesh tree scoped to this RU. Three tiers rendered as an SVG hierarchy: Gateway at root, Cluster Heads in mid layer, Sensor Nodes at leaves.

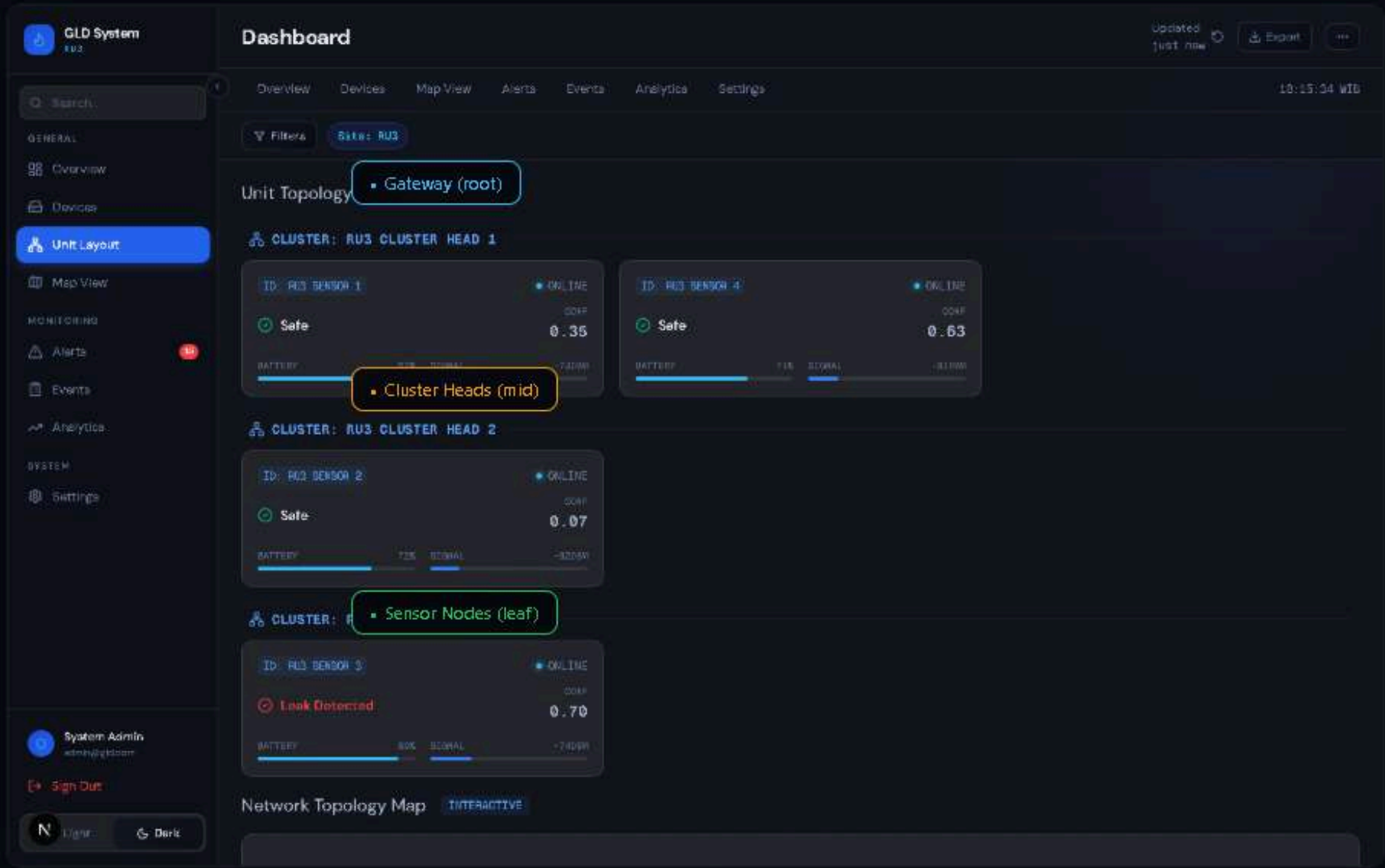
- ### Three-Tier Hierarchy

Gateway (1 per RU) → Cluster Heads (4 per RU) → GLD Sensor Nodes (up to 28 per RU).
- ### LoRa Dual-Radio

Radio A: 920 MHz SF7 STAR (GLD→CH). Radio B: 921 MHz SF9 MESH (CH→Gateway). Rendered as labelled edges.
- ### Color-Coded Status

Node fill: cyan = ONLINE, amber = WARNING, red = OFFLINE or active alarm. Click a node to select device.
- ### Live Device Count

Node labels show device name + last-seen health score. Click selects the device across all tabs.



Geospatial Map

Mapbox GL JS satellite view for this RU. Markers show health-colored device positions. Drag any marker to reposition — saved to the database instantly.

- Mapbox GL JS

Satellite / dark-v11 basemap with optional 3D building extrusions. Auto-fits to RU device bounds on load.
- Health-Colored Markers

Cyan = healthy, amber = warning, red = critical/offline. Pulsing ring on devices with active gas alerts.
- Supercluster Grouping

Markers auto-cluster at low zoom. Cluster bubble shows count; zoom in to expand individual pins.
- Drag-to-Reposition

Drag any pin to update device coordinates — mutations persisted via GraphQL, no page reload needed.

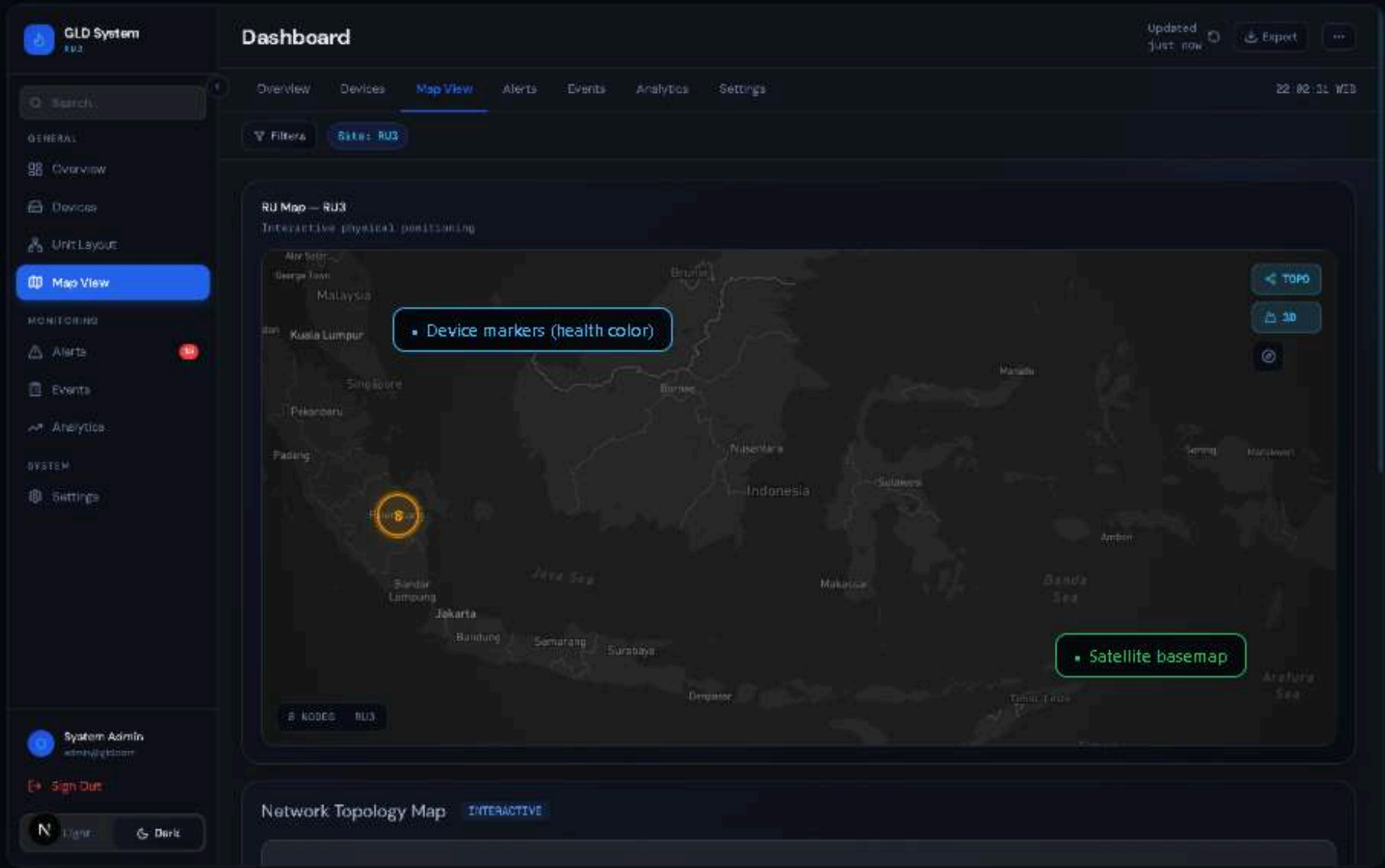
Mapbox GL JS

react-map-gl

supercluster

GraphQL

updateDevicePosition



MOCK-UP DASHBOARD MONITORING

Device Popup

Device Identity

AI Confidence + Risk

Battery · Signal · Health

Drag-to-Relocation

Alert Console

Live alarm feed filtered to this RU. Triggers when the AI model classifies a gas event with high confidence. Operators acknowledge and resolve alerts here.

**AI-Driven Triggers**

Alert fires when predicted_class ≠ 0 AND confidence ≥ HIGH threshold (default 0.80). 9 gas classes supported.

**Risk Levels**

HIGH: conf ≥ 0.80. MIDDLE: conf 0.70–0.79. LOW: below threshold. Badge color reflects severity.

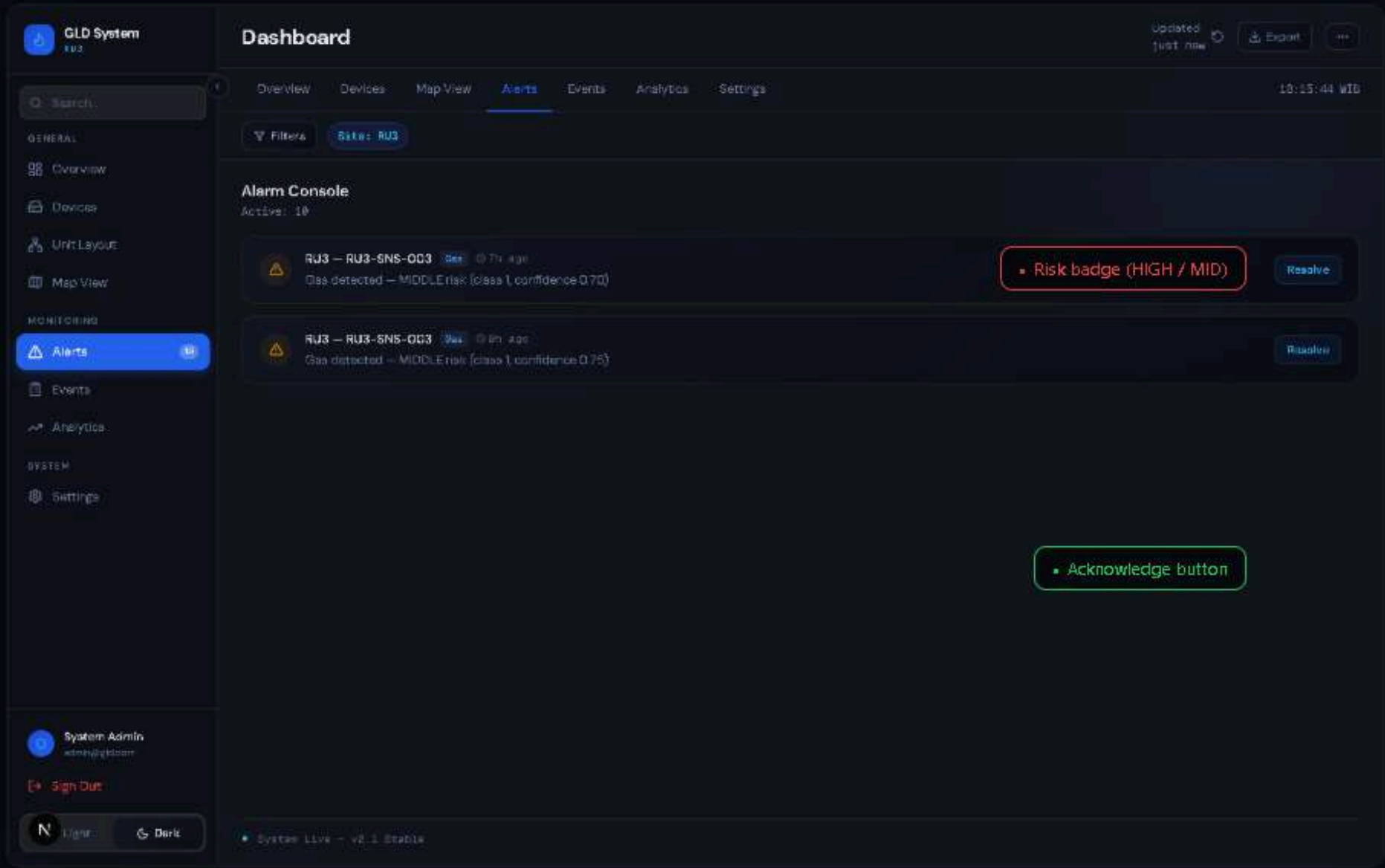
**Status Workflow**

ACTIVE → ACKNOWLEDGED → RESOLVED. Each transition logged in EventLog. Bell counter auto-decrements.

**Red Bell Badge**

Sidebar "Alerts" item shows live unresolved count. In this RU, only this unit's alerts are shown.

- AI classification
- GraphQL alerts
- EventLog mutation
- status transitions



Event Log

Immutable audit trail for compliance. Logs threshold breaches, device status changes, user logins, and config changes. Supports HTML export for incident reporting.

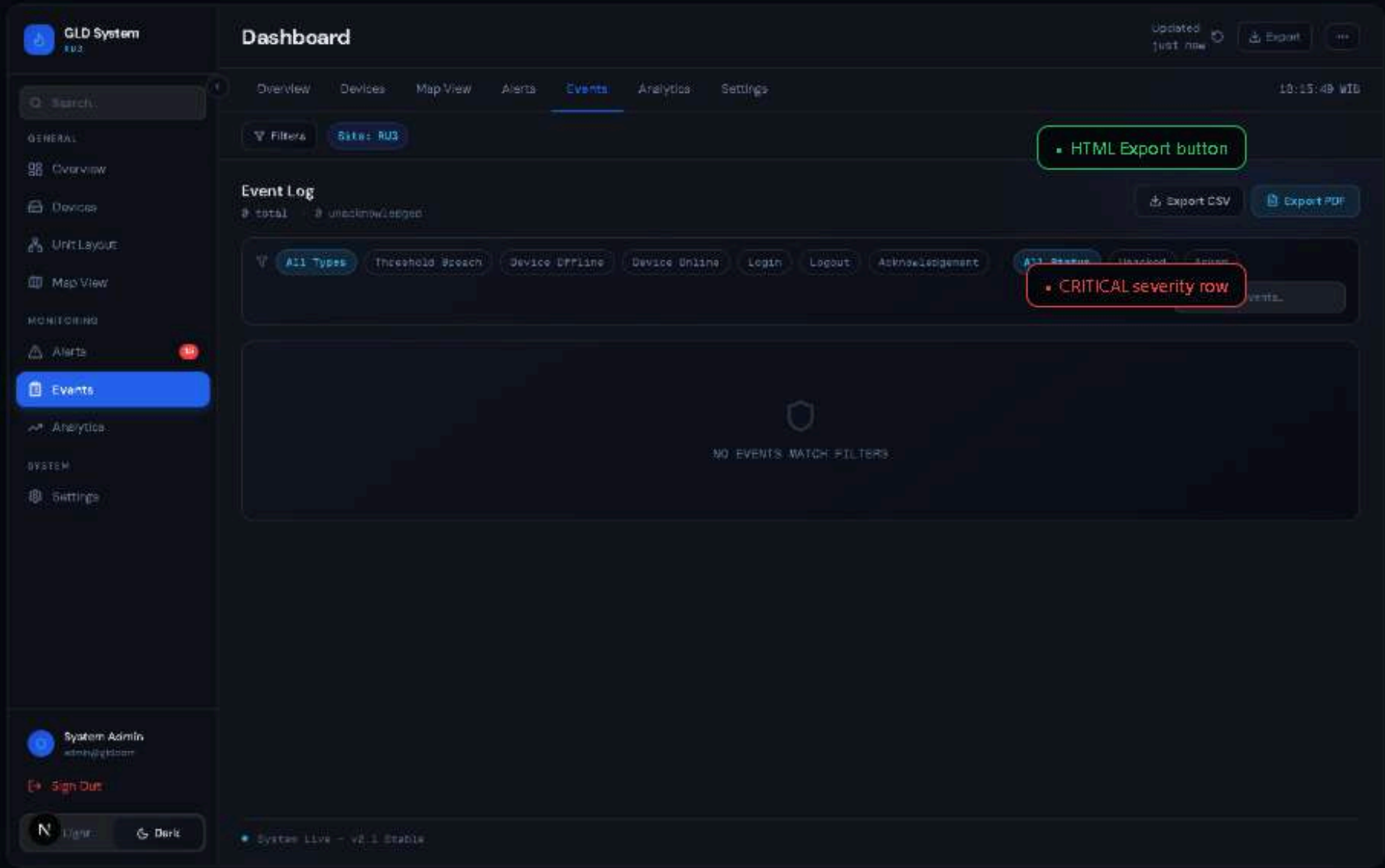
- Audit Trail**

Every GasReading crossing a threshold auto-logs THRESHOLD_BREACH. Device status flips log DEVICE_ONLINE/OFFLINE. 15-min dedup per device.
- Severity Levels**

INFO (blue) · WARNING (amber) · CRITICAL (red). HIGH risk reading → CRITICAL. MIDDLE → WARNING.
- HTML Export**

Download styled HTML report for compliance documentation, shift handover, or incident records.
- Filter & Sort**

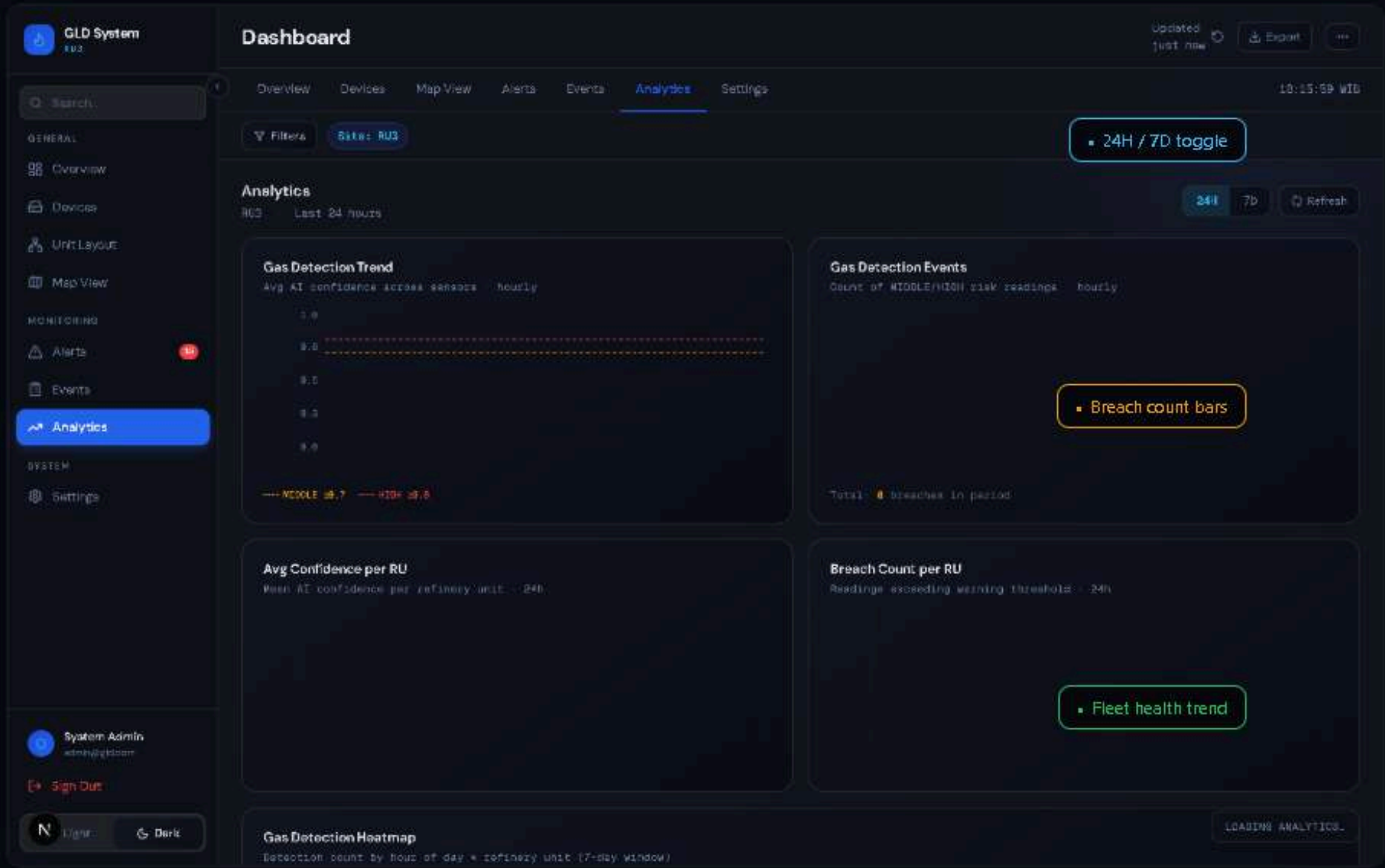
Filter by severity, date range, or device ID. Sorted newest-first by default.



Analytics Dashboard

Deeper trends for supervisors. Gas detection area chart, threshold breach bars, RU comparison, and fleet healthy/battery over time — all scoped to this unit.

- **Gas Detection Trend (Area)**
AI detection intensity over 24H or 7D. Reference lines at MIDDLE and HIGH thresholds. Flat = no event, spike = active gas.
- **Breach Count + RU Comparison**
Hourly/daily breach counts as bar chart. Horizontal bars compare avg detection level and breach frequency.
- **Gas Level Heatmap**
X = hour of day · Y = sensor row. Cell brightness = detection confidence. Reveals time-of-day leak patterns for inspection scheduling.
- **Fleet Health & Battery Trend**
Avg node health score and battery % over time. Downward trend predicts maintenance before a sensor goes offline mid-shift.



System Settings

Per-RU configuration panel. Set confidence thresholds, refresh interval, API URL, and manage local users — all persisted via GraphQL mutation to SQLite.

- 

MIDDLE Risk Threshold

AI confidence ≥ 0.70 AND class $\neq 0 \rightarrow$ MIDDLE risk.
Triggers WARNING event. Configurable slider.
- 

HIGH Risk Threshold

AI confidence ≥ 0.80 AND class $\neq 0 \rightarrow$ HIGH risk.
Triggers CRITICAL event + pushes to Alerts console.
- 

Refresh Interval

How often the dashboard polls for new readings and alerts. Default 10s; range 5–60s.
- 

User Management

Create / delete users scoped to this RU. Roles: ADMIN, OPERATOR, VIEWER. Actions logged to EventLog.



GLD System

RU2

Dashboard

Updated just now

Export

Search

GENERAL

Overview

Devices

Unit Layout

Map View

MONITORING

Alerts

Events

Analytics

SYSTEM

Settings

System Admin

admin@gld.com

Sign Out

Light

Dark

Overview

Devices

Map View

Alerts

Events

Analytics

Settings

Filters

Site: RU2

System Configuration

Global parameters for sensors and alerting logic.

Gas Alert Thresholds

MIDDLE Risk Threshold

confidence $\geq$ this $\rightarrow$ MIDDLE

0.70

HIGH Risk Threshold

confidence $\geq$ this $\rightarrow$ HIGH

0.80

Data Refresh Rate

Every 10 seconds (Balanced)

Save Configuration

HIGH threshold: ≥ 0.80

MIDDLE threshold: ≥ 0.70

Refresh rate control

User Management

Administrators and RU-specific operators

+ Add User

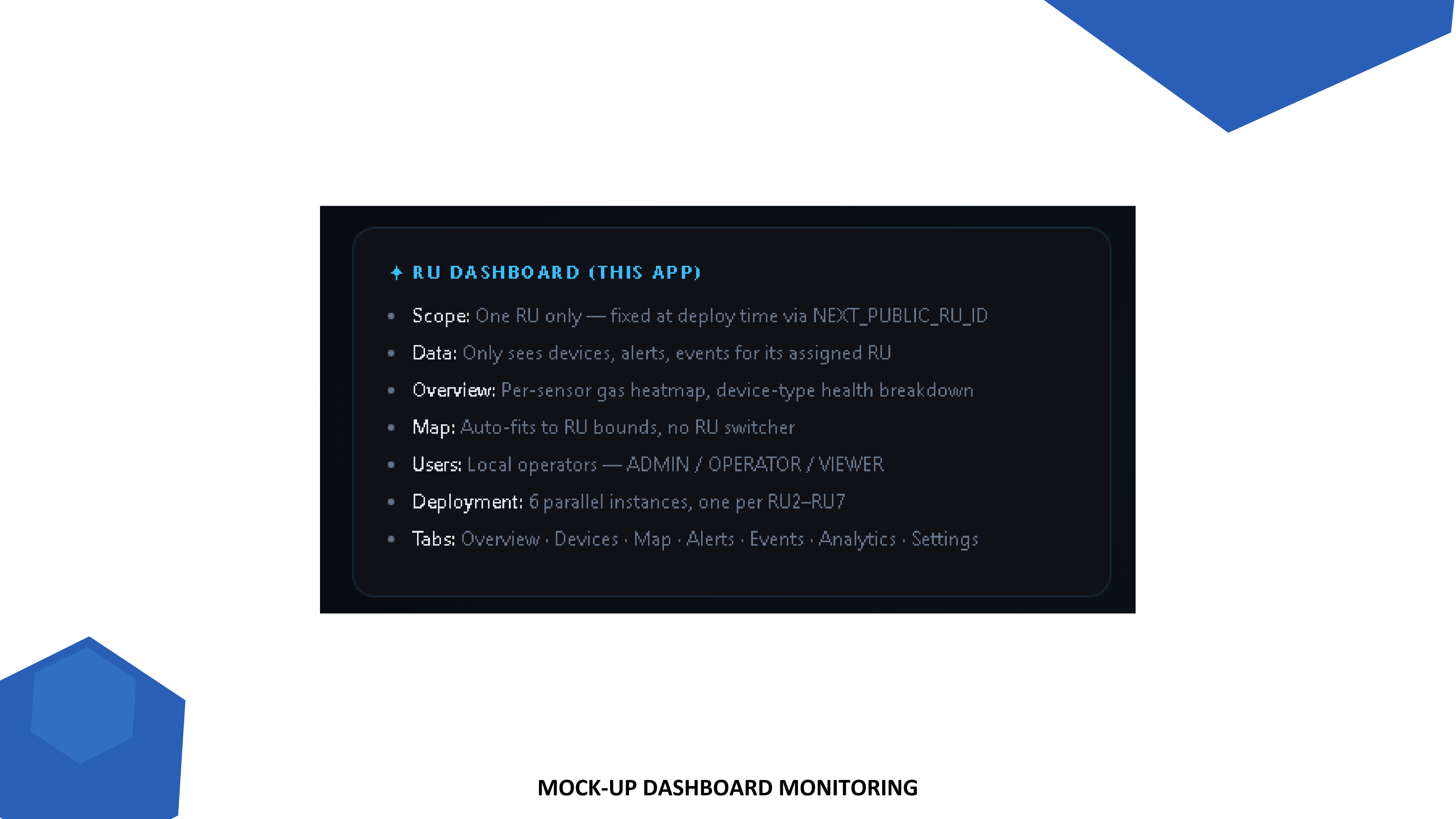
USER	ID	ADDRESS	ROLE	CREATED	ACTIONS
------	----	---------	------	---------	---------

System Live - v2.1 Stable

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MOCK-UP DASHBOARD MONITORING

SystemSettings model Prisma singleton
GraphQL updateSettings confidence thresholds



✦ RU DASHBOARD (THIS APP)

- Scope: One RU only — fixed at deploy time via NEXT_PUBLIC_RU_ID
- Data: Only sees devices, alerts, events for its assigned RU
- Overview: Per-sensor gas heatmap, device-type health breakdown
- Map: Auto-fits to RU bounds, no RU switcher
- Users: Local operators — ADMIN / OPERATOR / VIEWER
- Deployment: 6 parallel instances, one per RU2–RU7
- Tabs: Overview · Devices · Map · Alerts · Events · Analytics · Settings

NestJS Backend Architecture

Module Structure

```
AppModule
└─ DeviceModule
  │ DeviceResolver    + 12 GraphQL queries/mutations
  │ UserResolver      + 8 GraphQL queries/mutations
  │ DeviceService     + 21 business logic methods
  └─ PrismaService    + singleton DB client
```

— To be added —

```
MqttGatewayModule
└─ MqttGatewayService  subscribe + publish
└─ AppFrameParser      binary decode + CRC16
└─ SensorDataAlarmHandler
└─ ClusterBulkDataHandler
└─ ChHelloHandler
```

```
ReadingsPollerService + Node-RED MySQL staging
PullSchedulerService  + SERVER_PULL_REQUEST every 10s
```

Tech Stack

Runtime	NestJS 10 · Node.js
API	GraphQL (Apollo Server v4)
ORM	Prisma (SQLite dev / TimescaleDB prod)
MQTT	@nestjs/microservices + mqtt package
Port	4000

GraphQL API Surface

- `getDashboardStats()` — per-RU stats, alerts, fleet
- `getAnalytics(ruId, hours)` — trends, heatmap
- `sensorTimeline(ruId)` — hourly averages
- `eventLogs / acknowledgeEvent`
- `devices / createDevice / updateLocation`
- `login / createUser / deleteUser`

Prisma Schema & Planned Extensions

Current Schema (SQLite)

```
model Device {
  id          Int
  macAddress  String // "0x0005"
  name, deviceType, ruId
  location    Json    // {lat, lng}
  batteryStats, networkStats Json
  healthScore Float
  status      String
  parentId    Int?    // tree topology self-ref
}

model GasReading {
  id          Int
  deviceId    Int     // FK → Device
  ppm         Float   // legacy field
  timestamp   DateTime
}

model EventLog {
  type, severity, deviceId, ruId
  message, details Json
  acknowledged, ackNote
}
```

Fields to Add (Hardware Integration)

```
// GasReading additions
riskLevel  String // "LOW" | "MIDDLE" | "HIGH"
aiClass    Int    // 0-8
confidence Float  // 0.0-1.0
powerMode  String // "BATTERY" | "EXTERNAL"

// Device additions
batteryMv   Int?
uptimeS     Int?
chargerStatus String? // "CHARGING" | "FAULT"
meshDepth   Int?
lastSeenAt  DateTime?
clusterId   Int?
```

Production DB Upgrade

SQLite (dev) → **TimescaleDB/PostgreSQL** for time-series performance.
Target: 100M+ readings. Timing: before go-live.

Node-RED Hybrid Data Ingestion

Why Node-RED?

- Existing hardware already sends to MQTT / HTTP / Serial
- Acts as a normalize-and-buffer layer for mixed sources
- NestJS app logic stays clean — reads only from MySQL staging

Data Flow



Normalized Output Format

```
// MQTT source normalize
msg.payload = {
  mac_address: raw.mac || raw.macAddress,
  riskLevel:   raw.riskLevel, // "LOW"|"MIDDLE"|"HIGH"
  aiClass:     raw.aiClass,   // 0-8
  confidence:  raw.confidence, // 0.0-1.0
  source:      'mqtt'
};
```

Still Needed

- MySQL table schema (DESCRIBE sensor_readings;)
- Final field mapping from Node-RED to MySQL columns
- Device registration policy: skip unknown MAC or auto-create?

ReadingsPollerService (to build)

```
// Runs every SystemSettings.refreshInterval (10s)
SELECT * FROM sensor_readings
WHERE id > lastPolledId
ORDER BY id ASC LIMIT 100

// calls addReading() per row
// tracks lastPolledId for pagination
```

Frontend: Next.js 15 Dashboard

Tech Stack

LAYER	TECHNOLOGY
Framework	Next.js 15 App Router
Styling	Tailwind CSS (migration in progress)
Mapping	Mapbox GL JS + react-map-gl
Charts	Recharts
API	graphql-request → localhost:4000
Icons	Lucide React
State	React hooks + GraphQL queries (10s poll)

Key Dashboard Screens

- Live map view — all RU units with device markers
- Per-RU stats — alerts, battery, network health
- Event log — with severity filter + acknowledge
- Analytics — hourly trends, heatmap
- Device management — register, locate, rename
- Settings — thresholds, refresh interval

Performance Targets

METRIC	TARGET
MQTT → Map latency	< 1 SECOND
GraphQL query p95	< 100 MS
Device registration	< 5 SECONDS
Map load time	< 2 SECONDS
Battery alert response	< 500 MS
Concurrent devices	1,000 +
Concurrent users	500 +

Current Limitation

Frontend currently polls every 10s via GraphQL. Real-time Socketio upgrade is planned (Phase 1) to achieve the <1s alarm latency target.

Mapbox GL Layer Stack

- RU Boundary**
GeoJSON polygon per refinery unit
- Device Markers**
Drag-enabled · Health-colored icons · Click to inspect
- LoRa Mesh Topology Lines**
RSSI-based line thickness · CH → CH → GW routing path
- Thermal Plume Heatmap**
Animated gas concentration overlay · Updates on alarm
- Battery Heatmap**
Low-battery device clusters shown as red zones
- Alert Pins**
Severity-based marker size · Pulse animation on CRITICAL

Map Interaction Model

- Click device → side panel with readings + history
- Drag device marker → updates location in DB via GraphQL mutation
- Click alarm pin → opens event detail + acknowledge button
- RU boundary click → zooms to unit + shows RU stats

Planned: Interactive Mesh Topology

Drag-and-drop node rearrangement · Live RSSI display on edges · Parent failover visualization · Phase 3 feature

Planned: Thermal Camera Panel

Live feed embed on alarm · Automatic snapshot saved to event record · Link displayed in EventLog detail · Phase 3 feature

User Roles & Permissions

RU_Operator

Scope: Single RU - Read only

- ✓ View map & alerts
- ✓ Acknowledge events
- ✓ Monitor device status
- X Device registration
- X User management

RU_Supervisor

Scope: Single RU - Read + Device admin

- ✓ All Operator actions
- ✓ Register new devices
- ✓ Update device locations
- ✓ Export compliance reports
- X User management

RU_Admin

Scope: Single RU - Full unit control

- ✓ All Supervisor actions
- ✓ User management (RU)
- ✓ Maintenance scheduling
- ✓ Calibration tracking
- X Cross-RU analytics

Global_Admin

Scope: All RUs - Full system

- ✓ All Admin actions
- ✓ Cross-RU analytics
- ✓ System configuration
- ✓ ML model management
- ✓ Thresholds & settings

Current Auth State

Development mode: email-only login, no password check, no role enforcement. **Planned production:** Keycloak JWT with RU tenant guard — each token scoped to one or all RUs.

Event Log & Alarm Acknowledge Flow

EventLog Record Structure

FIELD	VALUES
type	THRESHOLD_BREACH, DEVICE_OFFLINE, BATTERY_LOW ...
severity	CRITICAL WARNING INFO
deviceId	FK to Device (optional)
ruid	Which refinery unit
message	Human-readable description
details	JSON — aiClass, confidence, batteryMv ...
acknowledged	Boolean
ackNote	Operator-entered notes

15-Minute Deduplication

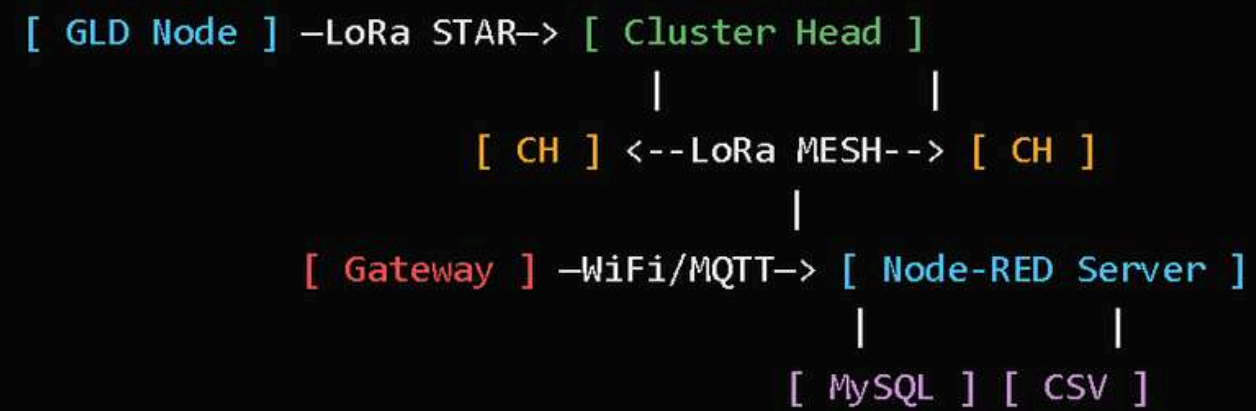
An alarm event is only created if no EventLog exists for the same device within the last 15 minutes. Prevents alert storms from continuously sensing nodes.

Alarm-to-ACK Flow



SYSTEM ARCHITECTURE

Arsitektur Sistem



[GLD Node] sensor endpoint • [Cluster Head] relay & cache • [Gateway] bridge • [Server] pemrosesan & storage

Perangkat Keras

GLD Node

- > ESP32-S3 (Xtensa LX7 dual-core)
- > 8× MQ sensor array
- > ADS1256 24-bit ADC
- > TCA9548A I2C multiplexer
- > RFM95W LoRa transceiver
- > Baterai + power management

Cluster Head & Gateway

- > ESP32-S3
- > SX1262 LoRa transceiver
- > CH: cache SRAM + alarm queue
- > Gateway: WiFi + MQTT client

Sensor Array

Sensor	Target Gas	Range
MQ-2	LPG, Propana, H ₂	300–10000 ppm
MQ-3	Alkohol, Etanol	25–500 ppm
MQ-4	CH ₄ (Metana)	300–10000 ppm
MQ-5	LPG, Alam Gas	300–10000 ppm
MQ-6	LPG, Propana	200–10000 ppm
MQ-7	CO (Karbon Monoksida)	20–2000 ppm
MQ-8	H ₂ (Hidrogen)	100–10000 ppm
MQ-135	NH ₃ , CO ₂ , asap	10–1000 ppm

MQ-2 MQ-3 MQ-4 MQ-5 MQ-6 MQ-7 MQ-8 MQ-135

Topologi Jaringan

LoRa STAR

- > GLD Nodes → Cluster Head
- > Point-to-multipoint
- > GLD tidak perlu routing
- > CH sebagai penerima tunggal

LoRa MESH

- > Cluster Heads ↔ Gateway
- > Multi-hop routing
- > Coverage area luas
- > Fallback jalur alternatif

```
F001 \ / CH1 -- CH2
F002 -- STAR -- CH3 <--MESH--> |
F003 / \ CH4 -- CH5 -- GW --TCP/MQTT-- Server
```

Alur Data End-to-End

1. **GLD Node:** Baca 8 sensor → ML inference → enkripsi AES-128-GCM → kirim via LoRa STAR
2. **Cluster Head:** Terima → cache record → siapkan relay MESH
3. **CH → Gateway (MESH):** Forward data via LoRa MESH multi-hop
4. **Gateway:** Terima MESH → parse frame → publish MQTT
5. **Server Node-RED:** Subscribe MQTT → dekripsi → dedup → simpan MySQL/CSV
6. **Alarm Push:** GLD deteksi gas → alarm flag → CH relay prioritas → server notifikasi

Sensor → ADC → ML → Encrypt → LoRa → CH → MESH → GW → MQTT → Node-RED → Decrypt → DB

GLD Node

Tugas

- > Membaca 8 sensor gas via ADS1256
- > Nulling & kalibrasi profil
- > Ekstraksi fitur (voltage, gain)
- > Inference ML (TFLite Micro)
- > Enkripsi payload AES-128-GCM
- > Transmisi LoRa STAR ke CH
- > Dataset generation via WiFi/MQTT

Firmware

v0.8.12 GLD

PlatformIO • ESP32-S3 • 240 MHz
16MB Flash • 8MB PSRAM
RadioLib • tfmicro

Mode Operasi

- > **Normal:** TX periodik ke CH
- > **Dataset:** Record streaming via WiFi
- > **Nulling:** Kalibrasi offset sensor
- > **Alarm:** Prioritas tinggi jika gas terdeteksi

ML Inference di Edge

TensorFlow Micro

- > Model neural network di ESP32-S3
- > Input: 16 fitur (8 voltage + 8 gain)
- > Output: gas class + confidence
- > Placeholder model terintegrasi
- > Siap diganti model hasil training

Pipeline

- > Baca sensor → raw ADC
- > Nulling correction
- > Scaler (standardisasi)
- > NeuralNetwork predict
- > Output: gasClass, confidence
- > Jika confidence > threshold → alarm

Status: Model placeholder aktif, inference pipeline terverifikasi

Cluster Head (CH)

Tugas

- > Menerima data dari GLD via LoRa STAR
- > Cache record GLD terbaru
- > Relay data ke Gateway via LoRa MESH
- > Alarm queue & prioritas forwarding
- > Menjawab pull request dari Gateway

Firmware

v0.7.1 CH

NodeCache • AlarmQueue • ChRuntime
STAR + MESH concurrent receive
Non-blocking runtime loop

Konfigurasi

ch1, ch2, ch3 — multiple CH units
MESH routing via hopList

Gateway

Tugas

- > Bridge LoRa MESH ↔ WiFi/MQTT
- > Terima data dari CH via MESH
- > Parse & validasi frame
- > Publish ke broker MQTT
- > Terima perintah dari server (pull request)
- > Kirim compact ACK ke CH

Firmware

v0.1.3 Gateway

GatewayMqttMeshMain • PubSubClient
MESH pull + forward logic
MQTT TLS/plain

Data Flow

MESH RX → parseStatus → MQTT publish
topic: gld/gateway/uplink

Server Node-RED

Tugas

- > Subscribe MQTT topics
- > Dekripsi payload AES-128-GCM
- > Deduplikasi record
- > Simpan ke MySQL & CSV
- > Alarm notification
- > Kirim perintah ke Gateway

Topik MQTT

gld/gateway/uplink	Data dari Gateway
gld/server/decoded	Hasil dekripsi
gld/server/alarm	Notifikasi alarm
gld/F001/cmd	Perintah ke GLD
gld/F001/dataset/#	Dataset streaming

MySQL gld_dataset table • CSV file logging • Aedes MQTT broker bench port 1884

Keamanan Data

AES-128-GCM

- > Enkripsi payload sensor di GLD Node
- > Dekripsi di server Node-RED
- > Key 128-bit, IV 12-byte random
- > Authentication tag (integrity check)
- > Tidak ada decrypt di CH atau Gateway

Format Payload

GCM_tag(16) + IV(12) + ciphertext

Frame header: srcId(2) + dstId(2) + seq(2) + flags(1) + ...

Protokol: v0.1.0

GLD Operator Lite

Fitur

- > Dashboard browser-based
- > Tab: Running, Dataset, Nulling, Log, Expert
- > Visualisasi real-time sensor
- > Perintah START/STOP dataset
- > Log streaming dari serial bridge

Teknologi

HTML5 + CSS3 + vanilla JavaScript
Python bridge (bridge.py)
Industrial dark theme

v0.2.0-lite-bridge

Status & Capaian

Selesai (12/12 Milestone)

- > Shared protocol & frame format
- > GLD board bring-up & nulling
- > STAR TX + STAR RX
- > CH runtime / cache / pull
- > Gateway MQTT + MESH bridge
- > Full normal path end-to-end
- > Alarm push live test
- > Dataset generation & recording
- > ML inference pipeline integrated

Versi Firmware

GLD	v0.8.12
CH	v0.7.1
Gateway	v0.1.3
Protokol	v0.1.0
Operator	v0.2.0

Host Tests

28/28 protocol tests passed

